

where k is a constant. If $y \neq 0$ we can write this equation in terms of differentials and integrate both sides as follows:

$$\int \frac{dy}{y} = \int k \, dt$$

$$\ln |y| = kt + C$$

$$|y| = e^{kt+C} = e^C e^{kt}$$

$$y = Ae^{kt}$$

The absolute value can be cleared by noting that we can write

$$y = e^C e^{kt} \quad \text{if } y > 0$$

$$y = -e^C e^{kt} \quad \text{if } y < 0$$

Therefore $y = Ae^{kt}$, where $A = \pm e^C$.

where $A (= \pm e^C)$ is an arbitrary constant. This is the solution presented in Sections 3.6 and 7.1. If $y = 0$ we cannot divide the differential equation by y . However, we can readily verify that, in this case, $y = 0$ is also a solution. Therefore the constant A in the solution $y = Ae^{kt}$ can also be 0. This corresponds to an equilibrium solution.

EXAMPLE 1

(a) Solve the differential equation $\frac{dy}{dx} = \frac{x^2}{y^2}$.

(b) Find the solution of this equation that satisfies the initial condition $y(0) = 2$.

SOLUTION

(a) We write the equation in terms of differentials and integrate both sides:

$$y^2 dy = x^2 dx$$

$$\int y^2 dy = \int x^2 dx$$

$$\frac{1}{3}y^3 = \frac{1}{3}x^3 + C$$

where C is an arbitrary constant. (We could have used a constant C_1 on the left side and another constant C_2 on the right side. But then we could combine these constants by writing $C = C_2 - C_1$.)

Solving for y , we get

$$y = \sqrt[3]{x^3 + 3C}$$

We could leave the solution like this or we could write it in the form

$$y = \sqrt[3]{x^3 + K}$$

where $K = 3C$. (Since C is an arbitrary constant, so is K .) Figure 1 plots this family of solutions.

(b) If we put $x = 0$ in the general solution in part (a), we get $y(0) = \sqrt[3]{K}$. To satisfy the initial condition $y(0) = 2$, we must have $\sqrt[3]{K} = 2$ and so $K = 8$. Thus the solution of the initial-value problem is

$$y = \sqrt[3]{x^3 + 8}$$

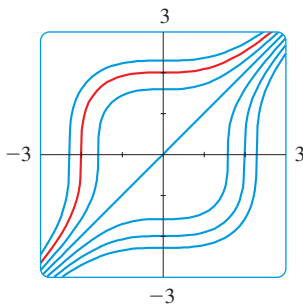


FIGURE 1

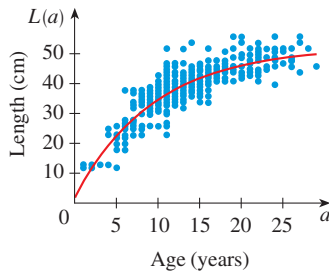
Graphs of several members of the family of solutions of the differential equation in Example 1. The solution of the initial-value problem in part (b) is shown in red.

EXAMPLE 2 | The von Bertalanffy growth equation A commonly used differential equation for the growth, in length, of an individual fish is

$$\frac{dL}{da} = k(L_\infty - L)$$

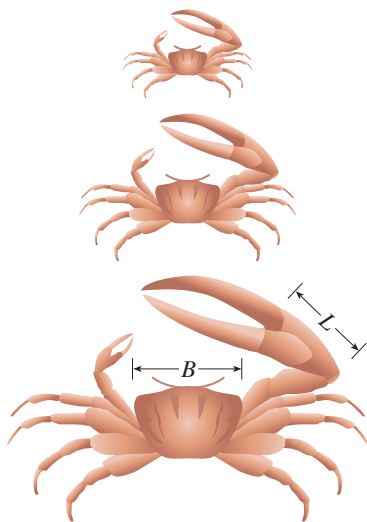
Von Bertalanffy

Ludwig von Bertalanffy (1901–1972) was an Austrian-born biologist who first published this differential equation for individual growth in 1934. It captures the idea that the rate of growth in length is proportional to the difference between current length and asymptotic length.

**FIGURE 2**

Age-length relationship for Atlantic redfish along with the solution to the von Bertalanffy equation with constants specific to redfish.

Source: Adapted from C. Stransky et al., “Age Determination and Growth of Atlantic Redfish (*Sebastes marinus* and *S. mentella*): Bias and Precision of Age Readers and Otolith Preparation Methods,” *ICES Journal of Marine Science* 62 (2005): 655–70.

**FIGURE 3**

Three crabs of different sizes, along with their claw lengths

where $L(a)$ is length (in cm) at age a (in years), L_∞ is the asymptotic length, and k is a positive constant whose units are 1/year.

- Find a family of solutions for length as a function of age.
- Find the solution that has an initial length of $L(0) = 2$.

SOLUTION

- Assuming $L \neq L_\infty$, we can write the equation in differential form as

$$\frac{dL}{L_\infty - L} = k da$$

Now integrate to obtain

$$\int \frac{dL}{L_\infty - L} = \int k da$$

or

$$-\ln|L_\infty - L| = ka + C_1$$

Now we can solve for L :

$$|L_\infty - L| = e^{-ka} e^{-C_1}$$

or

$$L = L_\infty - C e^{-ka}$$

where $C = \pm e^{-C_1}$ is an arbitrary constant. An example of this solution with particular constant values is shown in Figure 2.

If $L = L_\infty$, we cannot divide the differential equation by $L - L_\infty$, but we can verify that $L = L_\infty$ is itself another solution. Thus the constant C in the preceding solution can be 0 as well, and this again corresponds to an equilibrium solution.

- Setting $a = 0$ in the family of solutions from part (a) gives $L(0) = L_\infty - C$. To satisfy the initial condition $L(0) = 2$, we therefore require that $L_\infty - C = 2$, or $C = L_\infty - 2$. The desired solution is thus $L = L_\infty - (L_\infty - 2)e^{-ka}$ or

$$L = L_\infty(1 - e^{-ka}) + 2e^{-ka}$$

From this we can see why L_∞ is called the asymptotic length. As $a \rightarrow \infty$, $L \rightarrow L_\infty$. ■

EXAMPLE 3 | Allometric growth During growth, the claw of fiddler crabs increases in length at a *per unit rate* that is 1.57 times larger than that of its overall body width. In other words, if L and B denote claw length and body width, respectively (in mm), then

$$\frac{dL}{dt} \frac{1}{L} = 1.57 \frac{dB}{dt} \frac{1}{B}$$

(See Figure 3.) Find an equation that specifies claw length as a function of body width at any point during growth.

SOLUTION Multiplying both sides by dt gives

$$\frac{dL}{L} = 1.57 \frac{dB}{B}$$

Now integrate both sides to get

$$\ln L = 1.57 \ln B + C$$

On a log-log plot this is a straight line. Figure 4 plots data displaying this relationship. We can also rearrange our solution to the differential equation to obtain a power function for allometric scaling like those in Sections 1.2 and 1.5. Defining $k = e^C$, we obtain

$$L = kB^{1.57}$$

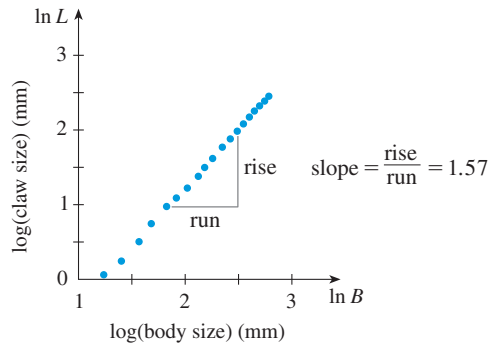


FIGURE 4

Data for the relationship between claw length and body width on a log-log plot

EXAMPLE 4 | Population dynamics Suppose the *per capita growth rate* of a population decreases as the population size n increases, in a way that is described by the expression $1/(1 + n)$. The differential equation for n is therefore

$$\frac{1}{n} \frac{dn}{dt} = \frac{1}{1 + n}$$

Solve this differential equation.

SOLUTION Writing the equation in differential form gives

$$\int \frac{1 + n}{n} dn = \int dt$$

(3)

$$\ln n + n = t + C$$

where C is an arbitrary constant. Equation 3 gives the family of solutions implicitly. In this case it's impossible to solve the equation to express n explicitly as a function of t .

The Gompertz differential equation assumes that the *per volume growth rate* of the tumor declines as the tumor volume gets larger according to the expression $a(\ln b - \ln V)$. Notice that the tumor growth rate is zero when $V = b$, where b represents the asymptotic tumor volume.

EXAMPLE 5 | Gompertz model of tumor growth The Gompertz differential equation models the growth of a tumor in volume V (in mm^3) and is given by

$$\frac{dV}{dt} = a(\ln b - \ln V)V$$

where a and b are positive constants.

- Find a family of solutions for tumor volume as a function of time.
- Find the solution that has an initial tumor volume of $V(0) = 1 \text{ mm}^3$.

SOLUTION First note that $\ln b - \ln V = \ln(b/V)$. Therefore, assuming $V \neq 0$ and $V \neq b$, we can write the equation in differential form and integrate as

$$\int \frac{dV}{V[\ln(b/V)]} = \int a dt$$

We can then integrate the left side using the substitution $u = \ln(b/V)$. We get

$$-\ln |\ln(b/V)| = at + C_1$$

Now we can solve for V by exponentiating both sides twice:

$$\ln(b/V) = Ce^{-at}$$

and then

$$b/V = e^{Ce^{-at}}$$

or

$$V = be^{-Ce^{-at}}$$

where $C = \pm e^{-C_1}$ is an arbitrary constant.

On the other hand, we can verify that $V = b$ is also an (equilibrium) solution.

(b) Setting $t = 0$ in the family of solutions from part (a) gives $V(0) = be^{-C}$. To satisfy the initial condition $V(0) = 1$, we therefore require that $1 = be^{-C}$ or $0 = \ln b - C$.

Therefore the desired solution is $V = be^{-(\ln b)e^{-at}}$ or

$$V = b(e^{-\ln b})e^{-at} \Rightarrow V = b\left(\frac{1}{b}\right)^{e^{-at}}$$

Figure 5 shows model predictions and data for three different sets of constant values with initial condition $V(0) = 35$.

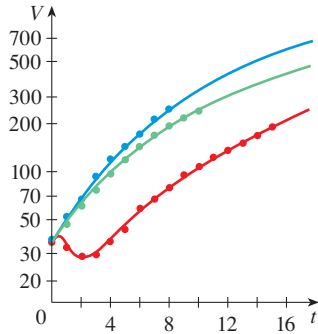


FIGURE 5

The solution of the Gompertz model fitted to tumor data.

Source: Adapted from D. Miklavčič et al., “Mathematical Modelling of Tumor Growth in Mice Following Electrotherapy and Bleomycin Treatment,” *Mathematics and Computers in Simulation* 39 (1995): 597–602.

EXAMPLE 6 | The logistic equation Find the solution to the following initial-value problem involving the logistic equation:

$$\frac{dN}{dt} = r\left(1 - \frac{N}{K}\right)N \quad N(0) = N_0$$

SOLUTION Assuming $N \neq 0$ and $N \neq K$, we can write the equation in differential form and integrate as

$$(4) \quad \int \frac{dN}{(1 - N/K)N} = \int r \, dt$$

To evaluate the integral on the left side, we write

$$\frac{1}{(1 - N/K)N} = \frac{K}{N(K - N)}$$

Using partial fractions (see Section 5.6), we get

$$\frac{K}{N(K - N)} = \frac{1}{N} + \frac{1}{K - N}$$

This enables us to rewrite Equation 4:

$$\int \left(\frac{1}{N} + \frac{1}{K - N} \right) dN = \int r dt$$

$$\ln |N| - \ln |K - N| = rt + C$$

$$\ln \left| \frac{K - N}{N} \right| = -rt - C$$

$$\left| \frac{K - N}{N} \right| = e^{-rt-C} = e^{-C} e^{-rt}$$

$$(5) \quad \frac{K - N}{N} = A e^{-rt}$$

where $A = \pm e^{-C}$. Solving Equation 5 for N , we get

$$\frac{K}{N} - 1 = A e^{-rt} \quad \Rightarrow \quad \frac{N}{K} = \frac{1}{1 + A e^{-rt}}$$

$$\text{so} \quad N = \frac{K}{1 + A e^{-rt}}$$

We find the value of A by putting $t = 0$ in Equation 5. If $t = 0$, then $N = N_0$ (the initial population), so

$$\frac{K - N_0}{N_0} = A e^0 = A$$

Thus the solution to the logistic equation is

$$N(t) = \frac{K}{1 + A e^{-rt}} \quad \text{where } A = \frac{K - N_0}{N_0}$$

On the other hand, if $N = 0$, then we can verify that this is also an (equilibrium) solution. Likewise, $N = K$ is an equilibrium solution.

We can now return to the model of yeast growth from page 424. As mentioned in Section 7.1, the model output in Figure 7.1.5 comes from the logistic growth equation with constant values $N_0 = 0.2$, $K = 210$, and $r = 0.55$. Substituting these values into the solution that we just obtained gives (after some rearrangement)

$$N(t) = \frac{42e^{0.55t}}{209.8 + 0.2e^{0.55t}}$$

This is exactly the solution presented on page 424. ■

EXERCISES 7.4

1–10 Solve the differential equation.

1. $\frac{dy}{dx} = xy^2$
2. $\frac{dy}{dx} = xe^{-y}$
3. $(x^2 + 1)y' = xy$
4. $(y^2 + xy^2)y' = 1$
5. $(y + \sin y)y' = x + x^3$
6. $\frac{du}{dr} = \frac{1 + \sqrt{r}}{1 + \sqrt{u}}$
7. $\frac{dy}{dt} = \frac{te^t}{y\sqrt{1 + y^2}}$
8. $\frac{dy}{d\theta} = \frac{e^y \sin^2 \theta}{y \sec \theta}$
9. $\frac{du}{dt} = 2 + 2u + t + tu$
10. $\frac{dz}{dt} + e^{t+z} = 0$

11–18 Find the solution of the differential equation that satisfies the given initial condition.

11. $\frac{dy}{dx} = \frac{x}{y}, \quad y(0) = -3$
12. $\frac{dy}{dx} = \frac{\ln x}{xy}, \quad y(1) = 2$
13. $\frac{du}{dt} = \frac{2t + \sec^2 t}{2u}, \quad u(0) = -5$
14. $y' = \frac{xy \sin x}{y + 1}, \quad y(0) = 1$
15. $x \ln x = y(1 + \sqrt{3 + y^2})y', \quad y(1) = 1$
16. $\frac{dP}{dt} = \sqrt{Pt}, \quad P(1) = 2$
17. $y' \tan x = a + y, \quad y(\pi/3) = a, \quad 0 < x < \pi/2$
18. $\frac{dL}{dt} = kL^2 \ln t, \quad L(1) = -1$

19. Find an equation of the curve that passes through the point $(0, 1)$ and whose slope at (x, y) is xy .
20. Find the function f such that $f'(x) = f(x)[1 - f(x)]$ and $f(0) = \frac{1}{2}$.
21. Solve the differential equation $y' = x + y$ by making the change of variable $u = x + y$.
22. Solve the differential equation $xy' = y + xe^{y/x}$ by making the change of variable $v = y/x$.
23. (a) Solve the differential equation $y' = 2x\sqrt{1 - y^2}$.
 (b) Solve the initial-value problem $y' = 2x\sqrt{1 - y^2}$, $y(0) = 0$, and graph the solution.
 (c) Does the initial-value problem $y' = 2x\sqrt{1 - y^2}$, $y(0) = 2$, have a solution? Explain.

 24. Solve the equation $e^{-y}y' + \cos x = 0$ and graph several members of the family of solutions. How does the solution curve change as the constant C varies?

 25. Solve the initial-value problem $y' = (\sin x)/\sin y$, $y(0) = \pi/2$, and graph the solution (if your CAS does implicit plots).

 26. Solve the equation $y' = x\sqrt{x^2 + 1}/(ye^y)$ and graph several members of the family of solutions (if your CAS does implicit plots). How does the solution curve change as the constant C varies?

 27–28

- (a) Use a computer algebra system to draw a direction field for the differential equation. Get a printout and use it to sketch some solution curves without solving the differential equation.
- (b) Solve the differential equation.
- (c) Use the CAS to draw several members of the family of solutions obtained in part (b). Compare with the curves from part (a).

27. $y' = y^2$

28. $y' = xy$

29–31 An **integral equation** is an equation that contains an unknown function $y(x)$ and an integral that involves $y(x)$. Solve the given integral equation. [Hint: Use an initial condition obtained from the integral equation.]

29. $y(x) = 2 + \int_2^x [t - ty(t)] dt$

30. $y(x) = 2 + \int_1^x \frac{dt}{ty(t)}, \quad x > 0$

31. $y(x) = 4 + \int_0^x 2t\sqrt{y(t)} dt$

32–34 Seasonality and habitat destruction The per capita growth rate of many species varies temporally for a variety of reasons, including seasonality and habitat destruction. Suppose $n(t)$ represents the population size at time t , where n is measured in individuals and t is measured in years. Solve the differential equation for habitat destruction and describe the predicted population dynamics.

32. $n' = e^{-t}n \quad n(0) = n_0$

Here the per capita growth rate declines over time, but always remains positive. It is modeled by the function e^{-t} .

33. $n' = (e^{-t} - 1)n \quad n(0) = n_0$

Here the per capita growth rate declines over time, starting at zero and becoming negative. It is modeled by the function $e^{-t} - 1$.

34. $n' = (r - at)n \quad n(0) = n_0$

Here the per capita growth rate declines over time, going

from positive to negative. It is modeled by the function $r - at$, where r and a are positive constants.

35. Noyes-Whitney drug dissolution Solve the initial-value problem in Exercise 7.1.14 for the Noyes-Whitney drug dissolution equation.

36. Weibull drug dissolution Solve the Weibull drug dissolution equation given in Exercise 7.1.15.

37–38 Bacteria colony growth In Exercises 1.6.35–36, we obtained difference equations for the growth of circular and spherical colonies of bacteria. These equations are based on the idea that nutrients for growth are available only at the colony–environment interface. Continuous-time versions of these equations are presented here, where k is a positive constant and n is the number of bacteria (in thousands). Solve each differential equation to find the size of the colony as a function of time. Assume $n(0) = 1$.

37. $\frac{dn}{dt} = kn^{1/2}$ (circular colony)

38. $\frac{dn}{dt} = kn^{2/3}$ (spherical colony)

39. Tumor growth The Gompertz equation in Example 5 is not the only possibility for modeling tumor growth. Suppose that a tumor can be modeled as a spherical collection of cells and it acquires resources for growth only through its surface area (like the spherical bacterial colony in Exercise 38). All cells in a tumor are also subject to a constant per capita death rate. The dynamics of tumor mass M (in grams) might therefore be modeled as

$$\frac{dM}{dt} = kM^{2/3} - \mu M$$

where μ and k are positive constants. The first term represents tumor growth via nutrients entering through the surface. The second term represents a constant per capita death rate.

- Assuming that $k = 1$ and $M(0) = 1$, find M as a function of t .
- What happens to the tumor mass as $t \rightarrow \infty$?
- Assuming tumor mass is proportional to its volume, the diameter of the tumor is related to its mass as $D = aM^{1/3}$, where $a > 0$. Derive a differential equation for D and show that it has the form of the von Bertalanffy equation in Example 2.

40. In an elementary chemical reaction, single molecules of two reactants A and B form a molecule of the product C: $A + B \rightarrow C$. The law of mass action states that the rate of reaction is proportional to the product of the concentrations of A and B:

$$\frac{d[C]}{dt} = k[A][B]$$

Thus, if the initial concentrations are $[A] = a$ moles/L and

$[B] = b$ moles/L and we write $x = [C]$, then we have

$$\frac{dx}{dt} = k(a - x)(b - x)$$

CAS

- Assuming that $a \neq b$, find x as a function of t . Use the fact that the initial concentration of C is 0.
- Find $x(t)$ assuming that $a = b$. How does this expression for $x(t)$ simplify if it is known that $[C] = \frac{1}{2}a$ after 20 seconds?

41. Population genetics Exercise 7.2.16 derives the following equation from population genetics that specifies the evolutionary dynamics of the frequency of a bacterial strain of interest:

$$\frac{dp}{dt} = sp(1 - p) \quad p(0) = p_0$$

where s is a constant. Find the solution, $p(t)$.

42. Mutation-selection balance The equation of Exercise 41 can be extended to account for a deleterious mutation that destroys the bacterial strain of interest. The differential equation becomes

$$\frac{dp}{dt} = sp(1 - p) - \mu p \quad p(0) = p_0$$

where μ is the mutation rate and $\mu > 0$ (see Exercise 7.2.19). Solve this initial-value problem for $s \neq \mu$.

43. Glucose administration A glucose solution is administered intravenously to the bloodstream at a constant rate r . As the glucose is added, it is converted into other substances and removed from the bloodstream at a rate that is proportional to the concentration at that time. Thus a model for the concentration $C(t)$ (in mg/mL) of the glucose solution in the bloodstream is

$$\frac{dC}{dt} = r - kC$$

where k is a positive constant.

- Suppose that the concentration at time $t = 0$ is C_0 . Determine the concentration at any time t by solving the initial-value problem.
- Assuming that $C_0 < r/k$, find $\lim_{t \rightarrow \infty} C(t)$ and interpret your answer.

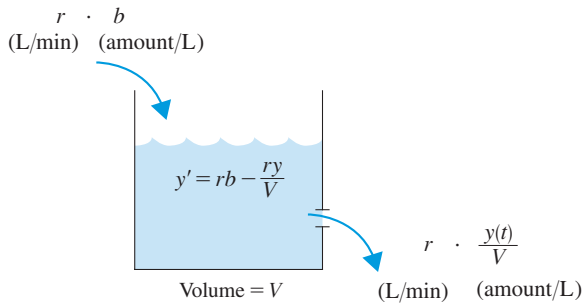
44. mRNA transcription The intermediate molecule mRNA arises in the decoding of DNA: it is produced by a process called transcription and it eventually decays. Suppose that the rate of transcription is changing exponentially according to the expression e^{bt} , where b is a positive constant and mRNA has a constant per capita decay rate of k . The number of mRNA transcript molecules T thus changes as

$$\frac{dT}{dt} = e^{bt} - kT$$

Although the form of this equation is similar to that from Exercise 43, the first term on the right side is now time-varying. As a result, the differential equation is no longer separable; however, the equation can be solved using the

change of variables $y(t) = e^{kt}T(t)$. Solve the differential equation using this technique.

45–48 Mixing problems Mixing problems arise in many areas of science. They typically involve a tank of fixed capacity filled with a well-mixed solution of some substance (such as salt). Solution of a given concentration enters the tank at a fixed rate and the mixture, thoroughly stirred, leaves at a fixed rate. We will focus on examples where the inflow and outflow rates are the same, so that the volume of solution in the tank remains constant. If $y(t)$ denotes the amount of substance in the tank at time t , then $y' = (\text{rate in}) - (\text{rate out})$.



- 45.** A tank contains 1000 L of brine with 15 kg of dissolved salt. Pure water enters the tank at a rate of 10 L/min. The solution is kept thoroughly mixed and drains from the tank at the same rate. How much salt is in the tank (a) after t minutes? (b) After 20 minutes?
- 46. Dialysis treatment** removes urea and other waste products from a patient's blood by diverting some of the blood flow externally through a machine called a dialyzer. Suppose that a patient's blood volume is V mL and blood is diverted through the dialyzer at a rate of K mL/min. At the start of treatment the patient's blood contains $c(0) = c_0$ mg/mL of urea.
- Formulate the process of dialysis as an initial-value problem.
 - What is the concentration of urea in the patient's blood after t minutes of dialysis? Compare your answer to Exercise 1.5.53.
- 47.** A vat with 500 gallons of beer contains 4% alcohol (by volume). Beer with 6% alcohol is pumped into the vat at a rate of 5 gal/min and the mixture is pumped out at the same rate. What is the percentage of alcohol after an hour?
- 48. Lung ventilation** A patient is placed on a ventilator to remove CO_2 from the lungs. Suppose that the rate of ventilation is 100 mL/s, with the percentage of CO_2 (by volume) in the inflow being zero. Suppose also that air is absorbed by the lungs at a rate of 10 mL/s and gas consisting of 100% CO_2 is excreted back into the lungs at the same rate. The volume of a typical pair of lungs is around 4000 mL. If the patient starts ventilation with 20% of lung volume being CO_2 , what volume of CO_2 will remain in the lungs after 30 minutes?
- 49.** When a raindrop falls, it increases in size and so its mass at time t is a function of t , namely, $m(t)$. The rate of growth of the mass is $km(t)$ for some positive constant k . When we apply Newton's Law of Motion to the raindrop, we get $(mv)' = gm$, where v is the velocity of the raindrop (directed downward) and g is the acceleration due to gravity. The terminal velocity of the raindrop is $\lim_{t \rightarrow \infty} v(t)$. Find an expression for the terminal velocity in terms of g and k .
- 50. Homeostasis** refers to a state in which the nutrient content of a consumer is independent of the nutrient content of its food. In the absence of homeostasis, a model proposed by Sterner and Elser is given by
- $$\frac{dy}{dx} = \frac{1}{\theta} \frac{y}{x}$$
- where x and y represent the nutrient content of the food and the consumer, respectively, and θ is a constant with $\theta \geq 1$.
- Solve the differential equation.
 - What happens when $\theta = 1$? What happens when $\theta \rightarrow \infty$?
- Source:* Adapted from R. Sterner et al., *Ecological Stoichiometry: The Biology of Elements from Molecules to the Biosphere* (Princeton, NJ: Princeton University Press, 2002).
- 51. Tissue culture** Let $A(t)$ be the area of a tissue culture at time t and let M be the final area of the tissue when growth is complete. Most cell divisions occur on the periphery of the tissue and the number of cells on the periphery is proportional to $\sqrt{A(t)}$. So a reasonable model for the growth of tissue is obtained by assuming that the rate of growth of the area is jointly proportional to $\sqrt{A(t)}$ and $M - A(t)$.
- Formulate a differential equation and use it to show that the tissue grows fastest when $A(t) = \frac{1}{3}M$.
 - Solve the differential equation to find an expression for $A(t)$. Use a computer algebra system to perform the integration.
- 52.** According to Newton's Law of Universal Gravitation, the gravitational force on an object of mass m that has been projected vertically upward from the earth's surface is
- $$F = \frac{mgR^2}{(x + R)^2}$$
- where $x = x(t)$ is the object's distance above the surface at time t , R is the earth's radius, and g is the acceleration due to gravity. Also, by Newton's Second Law, $F = ma = m(dv/dt)$ and so
- $$m \frac{dv}{dt} = -\frac{mgR^2}{(x + R)^2}$$
- Suppose a rocket is fired vertically upward with an initial velocity v_0 . Let h be the maximum height above the surface reached by the object. Show that
- $$v_0 = \sqrt{\frac{2gRh}{R + h}}$$
- [Hint: By the Chain Rule, $m(dv/dt) = mv(dv/dx)$.]

- (b) Calculate $v_e = \lim_{h \rightarrow \infty} v_0$. This limit is called the *escape velocity* for the earth.
- (c) Use $R = 3960$ mi and $g = 32$ ft/s² to calculate v_e in feet per second and in miles per second.

53. Species–area relationship The number of species found on an island typically increases with the area of the island.

Suppose that this relationship is such that the rate of increase with island area is always proportional to the density of species (that is, number of species per unit area) with a proportionality constant between 0 and 1. Find the function that describes the species-area relationship. Compare your answer to Example 1.5.14.

PROJECT Why Does Urea Concentration Rebound after Dialysis?

A patient undergoes dialysis treatment to remove urea from the bloodstream when the kidneys are not functioning properly. Blood is diverted from the patient through a machine that filters out the urea. In many patients, once a dialysis session ends there is a relatively rapid rebound in the concentration of urea in the blood—too rapid to be accounted for by the production of new urea (see Figure 1).

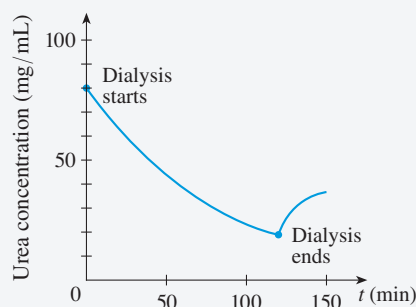


FIGURE 1

Urea rebound after dialysis

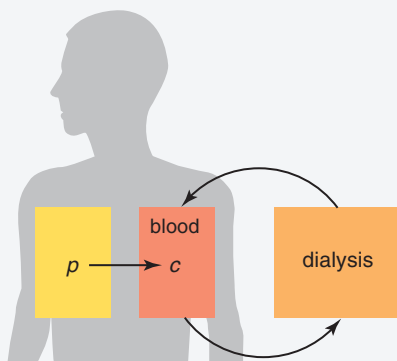


FIGURE 2

A schematic diagram of the two-compartment model

One explanation for this rebound is that urea also exists in other parts of the body, and there is continual movement of urea from these other areas into the bloodstream. Modeling this movement results in a so-called “two-compartment” model, as shown in Figure 2.

In Exercise 7.4.46 we saw that a common, one-compartment model for dialysis is

$$\frac{dc}{dt} = -\frac{K}{V}c$$

where K and V are positive constants and c is the concentration of urea in the blood (in mg/mL). To construct a two-compartment model we need to describe the dynamics using two variables, c for the concentration in the blood and p for the concentration in the inaccessible pool (both measured in mg/mL). A model for this process is

$$(1) \quad \frac{dc}{dt} = -\frac{K}{V}c + ap \quad \frac{dp}{dt} = -ap$$

where K , V , and a are positive constants.

1. Explain the terms in Equations 1 and the assumptions that underlie them.
2. The dynamics of c depend on both the concentration in the blood c and in the inaccessible pool p . However, the dynamics of p depend only on p and so we can solve the differential equation for p independently of the differential equation for c . What is this solution, assuming that the initial concentration of urea in the pool is c_0 ?

Nonautonomous differential equations of the form $y'(t) = f(t) - by(t)$ can be solved using the change of variables $x(t) = e^{bt}y(t)$. This is closely related to a technique for solving differential equations called “integrating factors.”

3. Use the solution you obtained in Problem 2, along with the equation for the dynamics of c in Equations 1, to write a single nonautonomous differential equation for the dynamics of c .
4. Suppose that the initial concentration of urea in the blood is also c_0 . Solve the initial-value problem for the concentration of urea in the blood as a function of time during dialysis using the change of variables described in the margin. Plot the concentration over time, assuming that $c_0 = 80$, $a = 0.015$, and $K/V = 0.03$.
5. Suppose dialysis treatment ends after 110 minutes. If we use $c^*(t)$ and $p^*(t)$ to denote the concentration of urea in the blood and in the inaccessible pool, t units of time after treatment has stopped, what is the initial-value problem for $c^*(t)$ and $p^*(t)$?
6. Solve the initial-value problem in Problem 5. Plot the solution along with that from Problem 4 on the same graph, using the constant values from Problem 4. What is the limiting value of the urea concentration in the blood after it has fully rebounded? Explain biologically why this limiting value occurs.

If there were flow of urea from the bloodstream back into the inaccessible pool, how do you think this would complicate the analysis of the model?

7.5 Systems of Differential Equations

The differential equations we have explored so far involve the dynamics of a single variable: population size, gene frequency, body size, and so on. Nevertheless, many biological phenomena—from the dynamics of interacting species to the dynamics of electrical impulses in a neuron—are described by multiple variables. This leads to models involving **coupled systems of differential equations**, meaning that the dynamics of each variable depend on the values of all variables in the system. This coupling means that we can’t solve one equation and then the other; we have to solve all equations simultaneously.

We begin our exploration of such models in this section and the next. A more complete treatment of coupled systems requires some background in multivariable calculus and is presented in Chapter 10. A brief introduction is included in this chapter, however, because it offers the flavor of the ideas and applications without requiring any further preparation. Here we focus on graphical techniques for *systems of two autonomous differential equations*. These techniques involve generalizing the ideas from Section 7.2 from plotting the dynamics along a *phase line* to plotting the dynamics in a *phase plane*. In order to do so, we first briefly consider the idea of parametric curves.

■ Parametric Curves

Imagine that a particle moves along the curve C shown in Figure 1. It is impossible to describe C by an equation of the form $y = f(x)$ because C fails the Vertical Line Test. But the x - and y -coordinates of the particle are functions of time and so we can write $x = f(t)$ and $y = g(t)$. Such a pair of equations is often a convenient way of describing a curve and gives rise to the following definition.

Suppose that x and y are both given as functions of a third variable t (called a **parameter**) by the equations

$$x = f(t) \quad y = g(t)$$

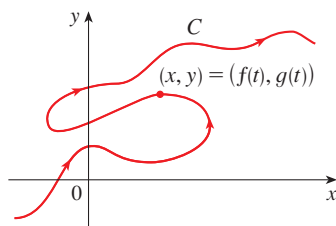


FIGURE 1

(called **parametric equations**). Each value of t determines a point (x, y) that we can plot in a coordinate plane. As t varies, the point $(x, y) = (f(t), g(t))$ varies and traces out a curve C , which we call a **parametric curve**. The parameter t does not necessarily represent time and, in fact, we could use a letter other than t for the parameter. But in the applications of parametric curves to systems of differential equations the parameter will be time and therefore we can interpret $(x, y) = (f(t), g(t))$ as the position of a particle at time t .

EXAMPLE 1 | Sketch and identify the curve defined by the parametric equations

$$x = t^2 - 2t \qquad y = t + 1$$

SOLUTION Each value of t gives a point on the curve, as shown in the table. For instance, if $t = 0$, then $x = 0, y = 1$ and so the corresponding point is $(0, 1)$. In Figure 2 we plot the points (x, y) determined by several values of the parameter t and we join them to produce a curve.

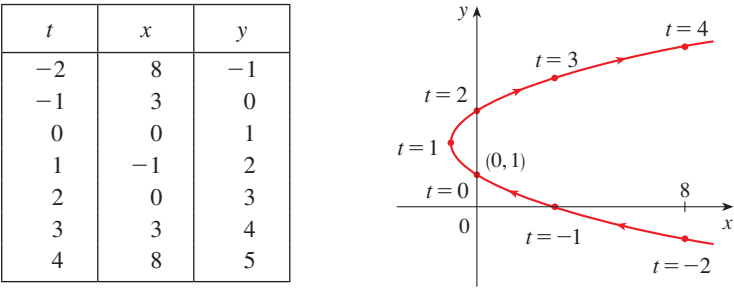


FIGURE 2

A particle whose position is given by the parametric equations moves along the curve in the direction of the arrows as t increases. Notice that the consecutive points marked on the curve appear at equal time intervals but not at equal distances. That is because the particle slows down and then speeds up as t increases.

It appears from Figure 2 that the curve traced out by the particle may be a parabola. This can be confirmed by eliminating the parameter t as follows. We obtain $t = y - 1$ from the second equation and substitute into the first equation. This gives

$$x = t^2 - 2t = (y - 1)^2 - 2(y - 1) = y^2 - 4y + 3$$

and so the curve represented by the given parametric equations is the parabola $x = y^2 - 4y + 3$. ■

The notion of a parametric curve is very useful in the study of systems of differential equations.

■ **Systems of Two Autonomous Differential Equations**

Consider a situation in which one species, called the *prey*, has an ample food supply and the second species, called the *predator*, feeds on the prey. Examples of prey and predators include rabbits and wolves in an isolated forest, small fish and sharks, aphids and ladybugs, and bacteria and amoebas. Our model will have two dependent variables, and both are functions of time. We let $R(t)$ be the number of prey (using R for rabbits) and $W(t)$ be the number of predators (with W for wolves) at time t .

This equation in x and y describes *where* the particle has been, but it doesn't tell us *when* the particle was at a particular point. The parametric equations have an advantage—they tell us *when* the particle was at a point. They also indicate the *direction* of the motion.

In the absence of predators, the ample food supply would support exponential growth of the prey, that is,

$$\frac{dR}{dt} = rR \quad \text{where } r \text{ is a positive constant}$$

In the absence of prey, we assume that the predator population would decline through mortality at a rate proportional to itself, that is,

$$\frac{dW}{dt} = -kW \quad \text{where } k \text{ is a positive constant}$$

With both species present, however, we assume that the principal cause of death among the prey is being eaten by a predator, and the birth rate of the predators depends on their available food supply, namely, the prey. We also assume that the two species encounter each other at a rate that is proportional to both populations and is therefore proportional to the product RW . This is referred to as the **principle of mass action**: the rate of encounter of two entities is proportional to the densities of each. A system of two *coupled* differential equations that incorporates these assumptions is as follows:

W represents the predator.
 R represents the prey.

$$(1) \quad \frac{dR}{dt} = rR - aRW \quad \frac{dW}{dt} = -kW + bRW$$

where k , r , a , and b are positive constants. Notice that the term $-aRW$ decreases the growth rate of the prey and the term bRW increases the growth rate of the predators.

The equations in (1) are known as the **predator-prey equations**, or the **Lotka-Volterra equations**. A **solution** of this system of equations is a pair of functions $R(t)$ and $W(t)$ that describe the populations of prey and predator as functions of time. Therefore a solution can be represented as a parametric curve $(x, y) = (R(t), W(t))$ in the plane. The graphical techniques developed here and in the next section are formulated in terms of such parametric curves.

The Lotka-Volterra equations were proposed as a model to explain the variations in the shark and food-fish populations in the Adriatic Sea by the Italian mathematician Vito Volterra (1860–1940).

EXAMPLE 2 | Lotka-Volterra equations Suppose that populations of rabbits and wolves are described by the Lotka-Volterra equations (1) with $r = 0.08$, $a = 0.001$, $k = 0.02$, and $b = 0.00002$. The time t is measured in months.

- Use the system of differential equations to find an expression for dW/dR .
- Draw a direction field for the resulting differential equation in the RW -plane. Then use that direction field to sketch some parametric curves representing solutions of Equations 1.
- Suppose that, at some point in time, there are 1000 rabbits and 40 wolves. Draw the corresponding parametric curve representing the solution for these initial conditions. Use it to describe the changes in both population levels.
- Use part (c) to make sketches of R and W as functions of t .

SOLUTION

- We use the Chain Rule to eliminate t :

$$\begin{aligned} \frac{dW}{dt} &= \frac{dW}{dR} \frac{dR}{dt} \\ \text{so} \quad \frac{dW}{dR} &= \frac{\frac{dW}{dt}}{\frac{dR}{dt}} = \frac{-0.02W + 0.00002RW}{0.08R - 0.001RW} \end{aligned}$$

(b) If we think of W as a function of R , we have the differential equation

$$\frac{dW}{dR} = \frac{-0.02W + 0.00002RW}{0.08R - 0.001RW}$$

We draw the direction field for this differential equation in Figure 3. This direction field is always tangent to the parametric curves representing solutions to Equations 1, as shown in Figure 4. If we move along a curve, we observe how the relationship between R and W changes as time passes. Although it is not obvious from the direction field, it can be shown that the curves are closed in the sense that if we travel along a curve, we always return to our starting point.

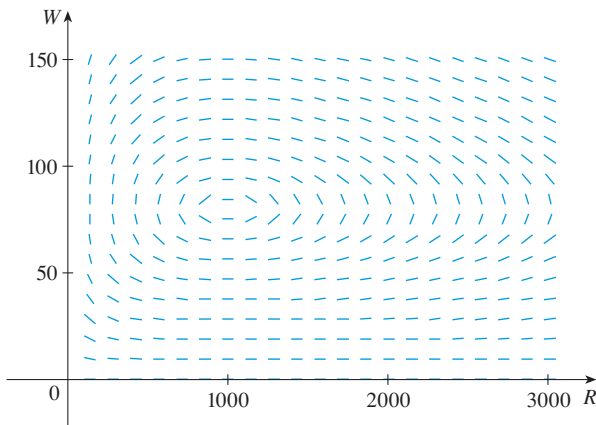


FIGURE 3 Direction field for the predator-prey system

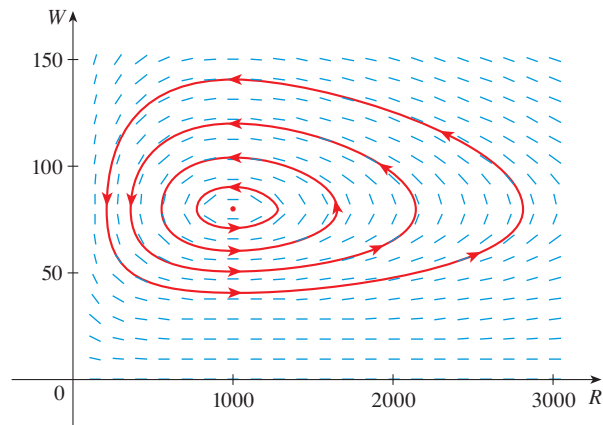


FIGURE 4 Phase portrait of the system

When we represent solutions of a system of differential equations as parametric curves in Figure 4, we refer to the RW -plane as the **phase plane**. This plane is the two-dimensional counterpart of the phase plots considered in Section 7.2. There, when we had a single variable, we plotted arrows on a line corresponding to the variable. Movement could be in either direction along the line and the arrows indicated the direction of this movement. Now, with two variables, we plot arrows in the plane corresponding to the two variables. Movement can now be in any direction in the plane and again the arrows indicate the direction of this movement. The parametric curves in the phase plane are called **phase trajectories**, and so a phase trajectory is a path traced out by solutions (R, W) as time goes by. A **phase portrait** consists of typical phase trajectories, as shown in Figure 4.

(c) Starting with 1000 rabbits and 40 wolves corresponds to drawing the parametric curve through the point $P_0(1000, 40)$. Figure 5 shows this phase trajectory with the direction field removed. Starting at the point P_0 at time $t = 0$ and letting t increase, do we move clockwise or counterclockwise around the phase trajectory? If we put $R = 1000$ and $W = 40$ in the first differential equation, we get

$$\frac{dR}{dt} = 0.08(1000) - 0.001(1000)(40) = 80 - 40 = 40$$

Since $dR/dt > 0$, we conclude that R is increasing at P_0 and so we move counter-clockwise around the phase trajectory.

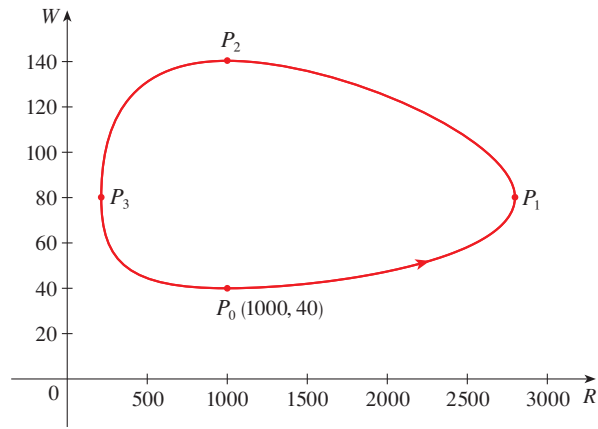


FIGURE 5
Phase trajectory through (1000, 40)

We see that at P_0 there aren't enough wolves to maintain a balance between the populations, so the rabbit population increases. That results in more wolves and eventually there are so many wolves that the rabbits have a hard time avoiding them. So the number of rabbits begins to decline (at P_1 , where we estimate that R reaches its maximum population of about 2800). This means that at some later time the wolf population starts to fall (at P_2 , where $R = 1000$ and $W \approx 140$). But this benefits the rabbits, so their population later starts to increase (at P_3 , where $W = 80$ and $R \approx 210$). As a consequence, the wolf population eventually starts to increase as well. This happens when the populations return to their initial values of $R = 1000$ and $W = 40$, and the entire cycle begins again.

(d) From the description in part (c) of how the rabbit and wolf populations rise and fall, we can sketch the graphs of $R(t)$ and $W(t)$. Suppose the points P_1 , P_2 , and P_3 in Figure 5 are reached at times t_1 , t_2 , and t_3 . Then we can sketch graphs of R and W as in Figure 6.

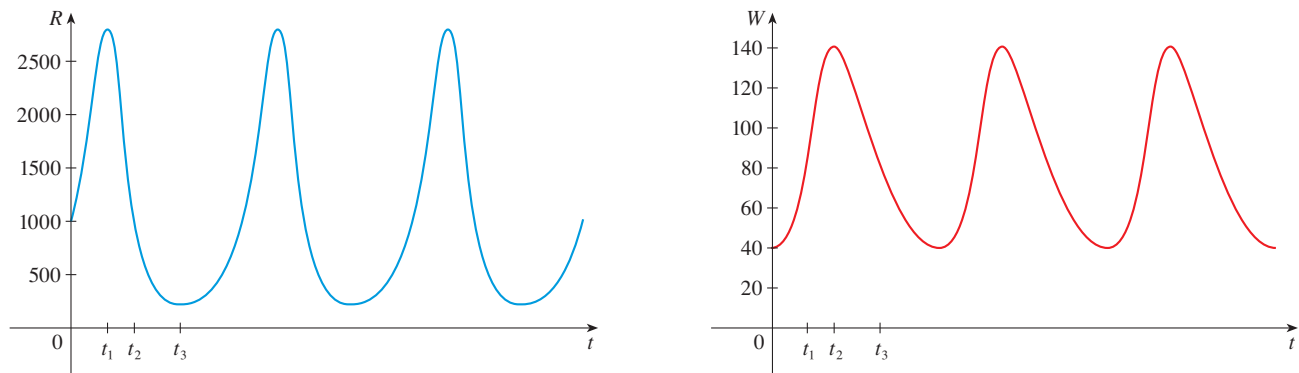


FIGURE 6 Graphs of the rabbit and wolf populations as functions of time

TEC In Module 7.5 you can change the coefficients in the Lotka-Volterra equations and observe the resulting changes in the phase trajectory and graphs of the rabbit and wolf populations.

To make the graphs easier to compare, we draw the graphs on the same axes but with different scales for R and W , as in Figure 7. Notice that the rabbits reach their maximum population size about a quarter of a cycle before the wolves.

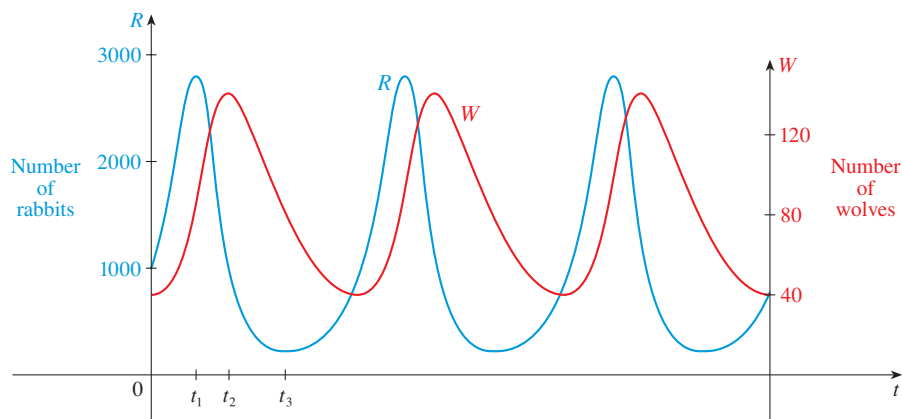


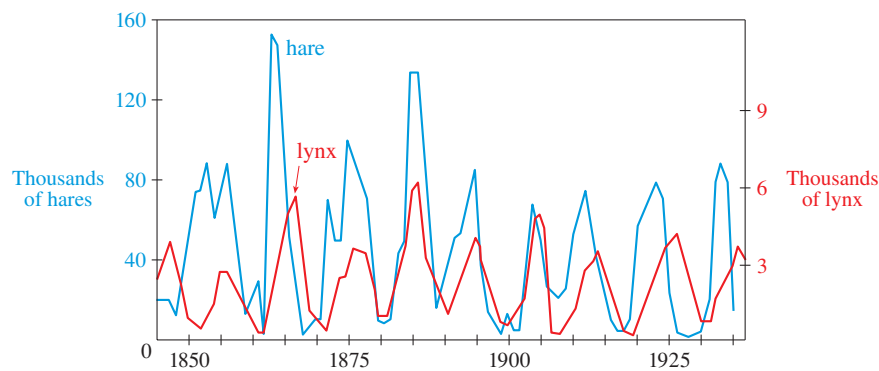
FIGURE 7
Comparison of the rabbit and wolf populations



Jeffrey Lepore / Science Source

FIGURE 8
Relative abundance of hare and lynx from Hudson's Bay Company records

An important part of the modeling process is to interpret our mathematical conclusions as real-world predictions and to test the predictions against real data. The Hudson's Bay Company, which started trading in animal furs in Canada in 1670, has kept records that date back to the 1840s. Figure 8 shows graphs of the number of pelts of the snowshoe hare and its predator, the Canada lynx, traded by the company over a 90-year period. You can see that the coupled oscillations in the hare and lynx populations predicted by the Lotka-Volterra model do actually occur and the period of these cycles is roughly 10 years.



Although the relatively simple Lotka-Volterra model has had some success in explaining and predicting coupled populations, more sophisticated models have also been proposed. One way to modify the Lotka-Volterra equations is to assume that, in the absence of predators, the prey grow according to a logistic model with carrying capacity M . Then the Lotka-Volterra equations (1) are replaced by the system of differential equations

$$\frac{dR}{dt} = rR \left(1 - \frac{R}{M} \right) - aRW \quad \frac{dW}{dt} = -kW + bRW$$

This model is investigated in Exercises 23 and 24.

Models have also been proposed to describe and predict population levels of two or more species that compete for the same resources or cooperate for mutual benefit. Such models are explored in Exercises 14 and 15.

EXERCISES 7.5

1–4 Sketch the curve by using the parametric equations to plot points. Indicate with an arrow the direction in which the curve is traced as t increases.

1. $x = t^2 + t$, $y = t^2 - t$, $-2 \leq t \leq 2$

2. $x = t^2$, $y = t^3 - 4t$, $-3 \leq t \leq 3$

3. $x = \cos^2 t$, $y = 1 - \sin t$, $0 \leq t \leq \pi/2$

4. $x = e^{-t} + t$, $y = e^t - t$, $-2 \leq t \leq 2$

5–8

(a) Sketch the curve by using the parametric equations to plot points. Indicate with an arrow the direction in which the curve is traced as t increases.

(b) Eliminate the parameter to find a Cartesian equation of the curve.

5. $x = 3t - 5$, $y = 2t + 1$

6. $x = 1 + 3t$, $y = 2 - t^2$

7. $x = \sqrt{t}$, $y = 1 - t$

8. $x = t^2$, $y = t^3$

9–12 Describe the motion of a particle with position (x, y) as t varies in the given interval.

9. $x = 3 + 2 \cos t$, $y = 1 + 2 \sin t$, $\pi/2 \leq t \leq 3\pi/2$

10. $x = 2 \sin t$, $y = 4 + \cos t$, $0 \leq t \leq 3\pi/2$

11. $x = 5 \sin t$, $y = 2 \cos t$, $-\pi \leq t \leq 5\pi$

12. $x = \sin t$, $y = \cos^2 t$, $-2\pi \leq t \leq 2\pi$

13. Predator-prey equations For each predator-prey system, determine which of the variables, x or y , represents the prey population and which represents the predator population. Is the growth of the prey restricted just by the predators or by other factors as well? Do the predators feed only on the prey or do they have additional food sources? Explain.

(a) $\frac{dx}{dt} = -0.05x + 0.0001xy$

$\frac{dy}{dt} = 0.1y - 0.005xy$

(b) $\frac{dx}{dt} = 0.2x - 0.0002x^2 - 0.006xy$

$\frac{dy}{dt} = -0.015y + 0.00008xy$

14. Competition and cooperation Each system of differential equations is a model for two species that either compete for the same resources or cooperate for mutual benefit (flowering plants and insect pollinators, for instance). Decide whether each system describes competition or

cooperation and explain why it is a reasonable model. (Ask yourself what effect an increase in one species has on the growth rate of the other.)

(a) $\frac{dx}{dt} = 0.12x - 0.0006x^2 + 0.00001xy$

$\frac{dy}{dt} = 0.08x + 0.00004xy$

(b) $\frac{dx}{dt} = 0.15x - 0.0002x^2 - 0.0006xy$

$\frac{dy}{dt} = 0.2y - 0.00008y^2 - 0.0002xy$

15. Cooperation, competition, or predation? The system of differential equations

$\frac{dx}{dt} = 0.5x - 0.004x^2 - 0.001xy$

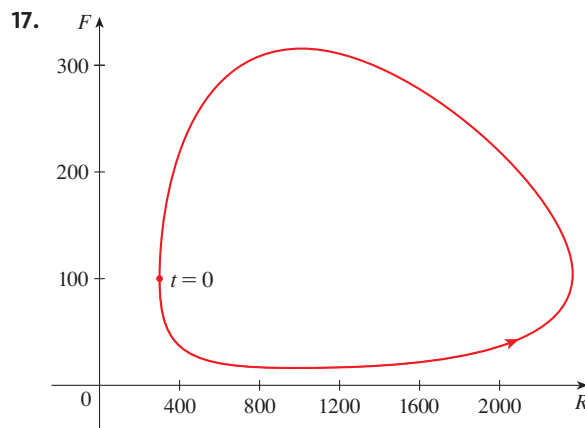
$\frac{dy}{dt} = 0.4y - 0.001y^2 - 0.002xy$

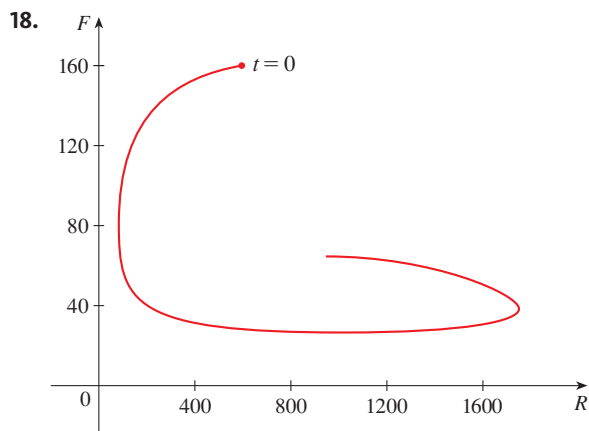
is a model for the populations of two species. Does the model describe cooperation, or competition, or a predator-prey relationship?

16. A food web Lynx eat snowshoe hares, and snowshoe hares eat woody plants, such as willows. Suppose that, in the absence of hares, the willow population will grow exponentially and the lynx population will decay exponentially. In the absence of lynx and willow, the hare population will decay exponentially. If $L(t)$, $H(t)$, and $W(t)$ represent the populations of these three species at time t , write a system of differential equations as a model for their dynamics. If the constants in your equation are all positive, explain why you have used plus or minus signs.

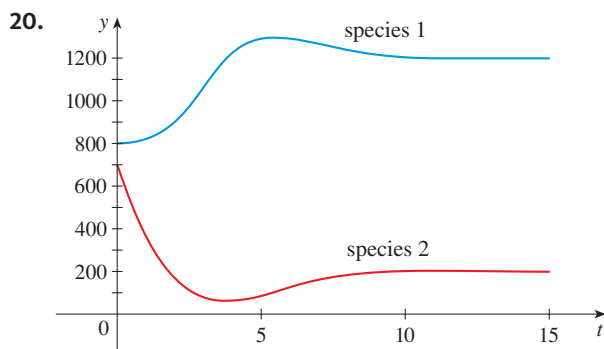
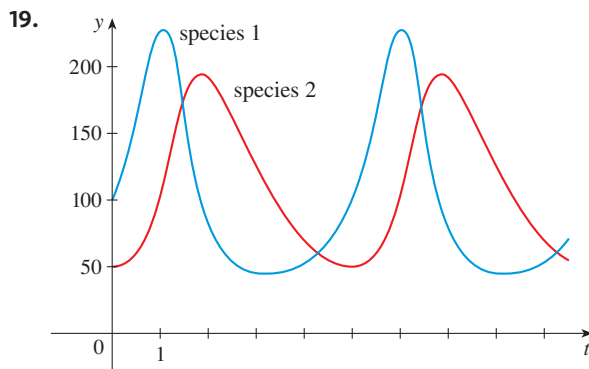
17–18 Rabbits and foxes A phase trajectory is shown for populations of rabbits (R) and foxes (F).

(a) Describe how each population changes as time goes by.
(b) Use your description to make a rough sketch of the graphs of R and F as functions of time.





19–20 Graphs of populations of two species are shown. Use them to sketch the corresponding phase trajectory.



21. Lotka-Volterra equations In Example 1(a) we showed that parametric curves describing the rabbit and wolf populations in the phase plane satisfy the differential equation

$$\frac{dW}{dR} = \frac{-0.02W + 0.00002RW}{0.08R - 0.001RW}$$

By solving this separable differential equation, show that

$$\frac{R^{0.02}W^{0.08}}{e^{0.00002R}e^{0.001W}} = C$$

where C is a constant.

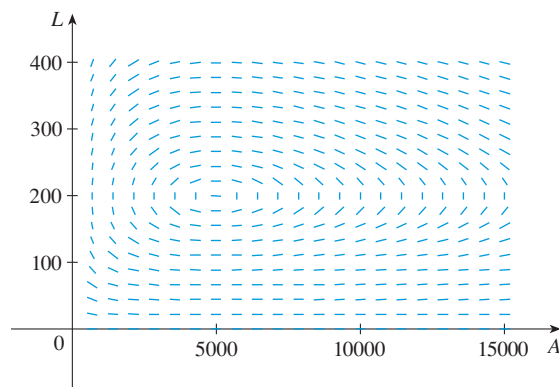
It is impossible to solve this equation for W as an explicit function of R (or vice versa). If you have a computer algebra system that graphs implicitly defined curves, use this equation and your CAS to draw the solution curve that passes through the point $(1000, 40)$ and compare with Figure 5.

22. Aphid-ladybug dynamics Populations of aphids and ladybugs are modeled by the equations

$$\frac{dA}{dt} = 2A - 0.01AL$$

$$\frac{dL}{dt} = -0.5L + 0.0001AL$$

- Find an expression for dL/dA .
- The direction field for the differential equation in part (b) is shown. Use it to sketch a phase portrait. What do the phase trajectories have in common?



- Suppose that at time $t = 0$ there are 1000 aphids and 200 ladybugs. Draw the corresponding phase trajectory and use it to describe how both populations change.
- Use part (c) to make rough sketches of the aphid and ladybug populations as functions of t . How are these two graphs related?

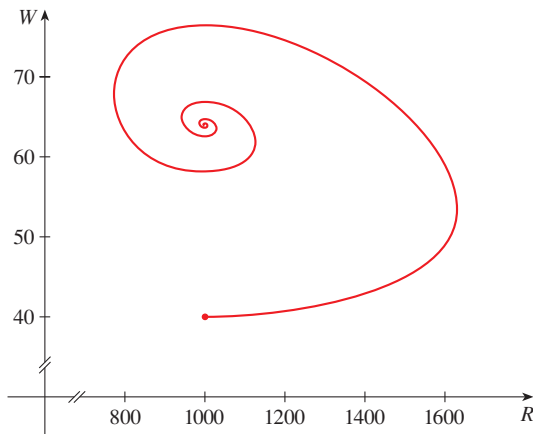
23. Modified predator-prey dynamics In Example 1 we used Lotka-Volterra equations to model populations of rabbits and wolves. Let's modify those equations as follows:

$$\frac{dR}{dt} = 0.08R(1 - 0.0002R) - 0.001RW$$

$$\frac{dW}{dt} = -0.02W + 0.00002RW$$

- According to these equations, what happens to the rabbit population in the absence of wolves?

- (b) The figure shows the phase trajectory that starts at the point (1000, 40). Describe what eventually happens to the rabbit and wolf populations.



- (c) Sketch graphs of the rabbit and wolf populations as functions of time.

24. Modified aphid-ladybug dynamics In Exercise 22 we modeled populations of aphids and ladybugs with a Lotka-Volterra system. Suppose we modify those equations as follows:

$$\frac{dA}{dt} = 2A(1 - 0.0001A) - 0.01AL$$

$$\frac{dL}{dt} = -0.5L + 0.0001AL$$

- (a) In the absence of ladybugs, what does the model predict about the aphids?
 (b) Find an expression for dL/dA .
 (c) Use a computer algebra system to draw a direction field for the differential equation in part (b). Then use the direction field to sketch a phase portrait. What do the phase trajectories have in common?
 (d) Suppose that at time $t = 0$ there are 1000 aphids and 200 ladybugs. Draw the corresponding phase trajectory and use it to describe how both populations change.
 (e) Use part (d) to make rough sketches of the aphid and ladybug populations as functions of t . How are these two graphs related?

CAS

PROJECT The Flight Path of Hunting Raptors



Many raptors, such as falcons and hawks, circle in on their prey while hunting rather than flying directly toward them (see Figure 1). One reason for this behavior is that they must aim one eye directly at the prey for maximum visual acuity and, because of the position of their eyes, this requires that they keep their direction of flight at a constant angle to the prey. Can we predict the flight path that a raptor will take toward its prey by describing this behavior mathematically?



FIGURE 1

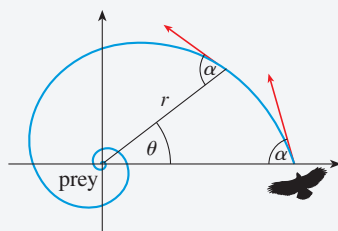


FIGURE 2

To simplify matters, let's consider the flight path in the horizontal plane only, under the assumption that the bird maintains its flight path at a constant angle α to the prey, where $0 < \alpha < \pi/2$ (see Figure 2). We need to describe the position of the bird in the plane at any time t . We could use the Cartesian coordinate system to do this, but the spiral structure of the flight path suggests a simpler approach: specify the distance r between the bird and its prey, along with the angle of rotation θ that identifies the location of the bird. Such coordinates are called polar coordinates.

The bird will circle the prey repeatedly as it closes in, meaning that the distance to the prey will decrease as the angle of rotation θ increases. To describe this process we

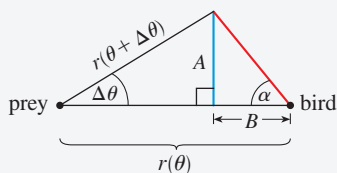


FIGURE 3

will therefore derive a differential equation for the distance to the prey r as a function of the angle of rotation θ .

1. In a small interval of rotation $\Delta\theta$ we can approximate the movement of the bird as a straight line (the red line in Figure 3). The initial distance to the prey is $r(\theta)$ and after the small change in θ it is $r(\theta + \Delta\theta)$. The known angles are also labeled in Figure 3. Our goal is to express $r(\theta + \Delta\theta)$ as a function of $r(\theta)$. To do this we first calculate some intermediate quantities. Show that the following relationships hold:

$$r(\theta + \Delta\theta) = \frac{r(\theta) - B}{\cos \Delta\theta} \quad B = \frac{\sin \Delta\theta}{\tan \alpha} r(\theta + \Delta\theta)$$

2. Use the relationships given in Problem 1 to obtain a formula for $r(\theta + \Delta\theta)$ in terms of $r(\theta)$, $\cos \Delta\theta$, $\sin \Delta\theta$, and $\tan \alpha$.
3. From your answer to Problem 2, show that the differential equation for r as a function of θ is given by

$$\frac{dr}{d\theta} = -r \cot \alpha$$

Hint: First obtain an expression for $\frac{r(\theta + \Delta\theta) - r(\theta)}{\Delta\theta}$ and then take the limit as $\Delta\theta \rightarrow 0$.

4. Suppose that we choose the initial angle of rotation to be zero when the prey is first spotted and the prey is a distance $r(0) = r_0$ from the bird. Solve the initial-value problem corresponding to the differential equation from Problem 3 with this initial condition. What is the distance to the prey as a function of the angle of rotation?
5. By what factor does the distance to the prey get reduced every time the bird circles the prey? Explain your answer.
6. Express the curve found in Problem 4 in Cartesian coordinates by specifying it as a parametric curve in x and y .



FIGURE 4

The curve describing the flight path of a hunting raptor found in Problem 4 is known as a **logarithmic spiral**. Its defining feature is that the tangent to the spiral at any point is always at a constant angle from a radial line joining the center of the spiral to this point. This means that the local geometry of the curve remains fixed and therefore it retains its shape no matter how big the spiral becomes. Perhaps because of this feature of maintaining a constant shape, logarithmic spirals are quite common in living organisms. One of the most famous examples is the spiral shell of a nautilus shown in Figure 4.

7.6 Phase Plane Analysis

The preceding section illustrated how the the phase plots from Section 7.2 can be extended to systems of two differential equations. These plots provide important information about the dynamics of the system but they can be tedious to construct in the absence of a computer. In this section we develop some tools to obtain a very general

qualitative understanding of the dynamics of *systems of two autonomous differential equations*. The technique—referred to as phase plane analysis—involves identifying equilibria of the equations and then determining the qualitative nature of the dynamics around these equilibria.

■ Equilibria

Recall the predator-prey equations from Section 7.5 for the population sizes of rabbits and wolves:

$$(1) \quad \frac{dR}{dt} = rR - aRW \quad \frac{dW}{dt} = -kW + bRW$$

The corresponding phase plane is shown in Figure 1. An *equilibrium* of this system of differential equations is a constant population size of rabbits $\hat{R}$ and of wolves $\hat{W}$ at which no further change in either occurs. This requires that both $dR/dt = 0$ and $dW/dt = 0$. Using Equations 1, this gives two equations in two unknowns: $rR - aRW = 0$ and $-kW + bRW = 0$.

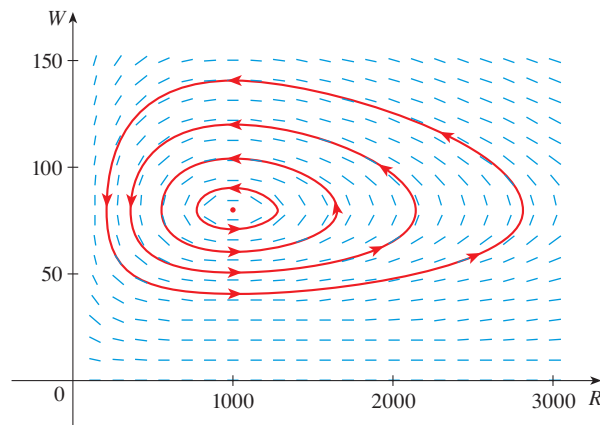


FIGURE 1
The predator-prey phase plot
when $r = 0.08$, $a = 0.001$,
 $k = 0.02$, and $b = 0.00002$

Definition Consider the autonomous system of differential equations

$$(2) \quad \frac{dx}{dt} = f(x, y) \quad \frac{dy}{dt} = g(x, y)$$

An **equilibrium** is a pair of values $(\hat{x}, \hat{y})$ such that both $dx/dt = 0$ and $dy/dt = 0$ when $x = \hat{x}$ and $y = \hat{y}$. This gives a pair of equations $f(\hat{x}, \hat{y}) = 0$ and $g(\hat{x}, \hat{y}) = 0$ that define the values of $\hat{x}$ and $\hat{y}$.

We can connect the pair of equations defining the equilibria of the predator-prey model to the phase plane in Figure 1. The equation $rR - aRW = 0$ must hold if the population size of rabbits is to remain constant; it can be factored to give $R(r - aW) = 0$. Therefore the population size of rabbits will remain constant if either $R = 0$ or $W = r/a$. The first of these equations corresponds to absence of rabbits altogether. The second corresponds to the population size of wolves at which the birth rate of rabbits is

exactly balanced by their death rate through predation. The lines defined by these equations are called the *R-nullclines* and are plotted on the phase plane shown in Figure 2(a).

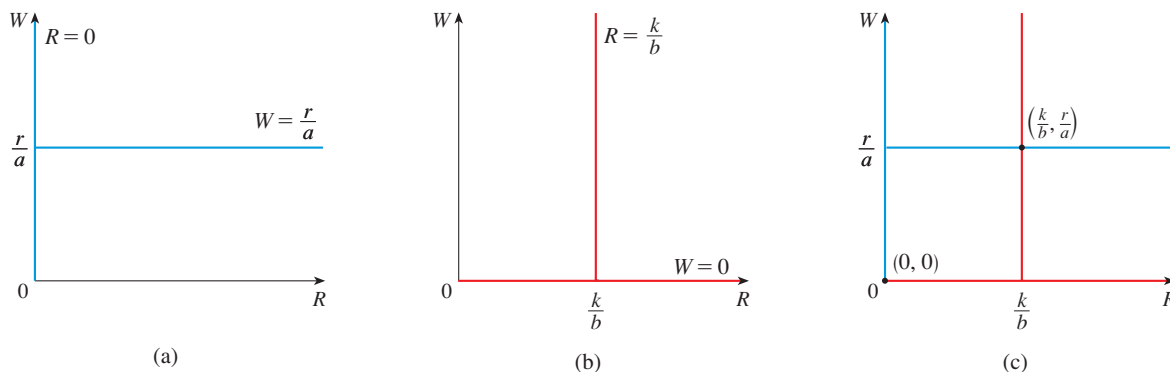


FIGURE 2

R -nullclines are blue, W -nullclines are red, and equilibria are black dots.

The second equation, $-kW + bRW = 0$, must hold if the population size of wolves is to remain constant; it can be factored and solved to give $W = 0$ and $R = k/b$. The first of these equations corresponds to absence of wolves. The second corresponds to the population size of rabbits at which the birth rate of wolves (through consumption of rabbits) is exactly balanced by their death rate. The lines defined by these equations are the *W-nullclines* and are plotted on the phase plane shown in Figure 2(b). Figure 2(c) shows the R - and W -nullclines plotted together.

Definition The **x -nullclines** of differential equations (2) are the curves in the xy -plane that satisfy the equation $f(x, y) = 0$. Along these curves, $dx/dt = 0$. The **y -nullclines** of differential equations (2) are the curves in the xy -plane that satisfy the equation $g(x, y) = 0$. Along these curves, $dy/dt = 0$.

Whenever a trajectory in the phase plane crosses a nullcline, it must do so either horizontally or vertically, depending on the nullcline in question. This is because movement in either the vertical or horizontal direction is zero on a nullcline since either $dy/dt = 0$ or $dx/dt = 0$.

Nullclines of the differential equations provide a graphical way to visualize the equilibria of differential equations. For example, in the predator-prey model, an equilibrium requires that *both* the predator and the prey population sizes remain constant through time. Geometrically, an equilibrium will therefore occur at any point where an R -nullcline intersects a W -nullcline. It will be precisely at such intersection points that both variables remain constant through time [see Figure 2(c)].

Finding Equilibria Graphically For differential equations (2) any point at which an x -nullcline intersects a y -nullcline is an equilibrium.

In addition to this visualization of equilibrium points and nullclines, we can sometimes derive expressions for the equilibria algebraically. The predator-prey model has two equilibria: (i) $\hat{R} = 0$, $\hat{W} = 0$ and (ii) $\hat{R} = k/b$, $\hat{W} = r/a$.

EXAMPLE 1 | Lotka-Volterra competition equations The differential equation for logistic population growth (Equation 7.1.4) can be extended to model competitive interactions between two species. Let's use $N_1(t)$ and $N_2(t)$ to denote the population size of species 1 and 2 at time t . Suppose that the per capita growth rate of

each species decreases linearly with the population size of each species. Specifically, the per capita growth rate of species 1 is

$$r \left(1 - \frac{N_1 + \alpha N_2}{K_1} \right)$$

where α , r , and K_1 are positive constants. Likewise, the per capita growth rate of species 2 is

$$r \left(1 - \frac{N_2 + \beta N_1}{K_2} \right)$$

where β and K_2 are positive constants. (Compare these per capita growth rates to the per capita growth rate in Equation 7.1.4.) This gives the system

$$(3) \quad \frac{dN_1}{dt} = r \left(1 - \frac{N_1 + \alpha N_2}{K_1} \right) N_1 \quad \frac{dN_2}{dt} = r \left(1 - \frac{N_2 + \beta N_1}{K_2} \right) N_2$$

- (a) Suppose $r = 1$, $K_1 = 1000$, $K_2 = 600$, $\alpha = 2$, and $\beta = 1$. Find the N_1 - and N_2 -nullclines and plot them on the phase plane. Indicate all equilibria.
 (b) Calculate the equilibria algebraically.
 (c) Suppose instead that $\beta = 0$, but all other constants have the same values as in part (a). Calculate the equilibria algebraically.

SOLUTION

- (a) The N_1 -nullclines satisfy $dN_1/dt = 0$ or

$$\left(1 - \frac{N_1 + 2N_2}{1000} \right) N_1 = 0$$

Therefore the N_1 -nullclines are $N_1 = 0$ and $N_1 + 2N_2 = 1000$. The second equation can be rewritten as $N_2 = 500 - \frac{1}{2}N_1$. It is plotted in Figure 3(a). The N_2 -nullclines satisfy $dN_2/dt = 0$ or

$$\left(1 - \frac{N_2 + N_1}{600} \right) N_2 = 0$$

The N_2 -nullclines are therefore $N_2 = 0$ and $N_1 + N_2 = 600$. The second equation can be rewritten as $N_2 = 600 - N_1$ and is plotted in Figure 3(b). The nullclines are plotted together, along with the equilibria, in Figure 3(c).

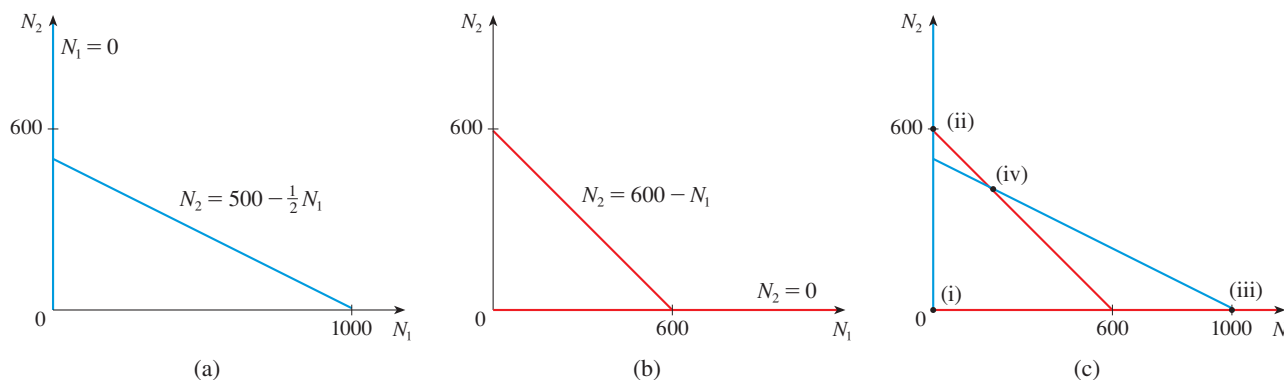


FIGURE 3 N_1 -nullclines are blue, N_2 -nullclines are red, and equilibria are black dots.

(b) Equilibria are pairs of values $(\hat{N}_1, \hat{N}_2)$ that simultaneously satisfy the pair of equations

$$\left(1 - \frac{\hat{N}_1 + 2\hat{N}_2}{1000}\right)\hat{N}_1 = 0 \quad \text{and} \quad \left(1 - \frac{\hat{N}_2 + \hat{N}_1}{600}\right)\hat{N}_2 = 0$$

We can calculate the equilibria by solving the first equation for $\hat{N}_1$, substituting the result into the second equation, and then solving it for $\hat{N}_2$. There are two solutions to the first equation: $\hat{N}_1 = 0$ and $\hat{N}_1 = 1000 - 2\hat{N}_2$. We consider each of these in turn.

Substituting $\hat{N}_1 = 0$ into the second equation gives

$$\left(1 - \frac{\hat{N}_2}{600}\right)\hat{N}_2 = 0$$

Solving for $\hat{N}_2$ gives $\hat{N}_2 = 0$ and $\hat{N}_2 = 600$. Therefore two equilibria are (i) $\hat{N}_1 = 0$, $\hat{N}_2 = 0$ and (ii) $\hat{N}_1 = 0$, $\hat{N}_2 = 600$ [see Figure 3(c)].

Substituting $\hat{N}_1 = 1000 - 2\hat{N}_2$ into the second equation gives

$$\left(1 - \frac{\hat{N}_2 + (1000 - 2\hat{N}_2)}{600}\right)\hat{N}_2 = 0$$

Solving for $\hat{N}_2$ gives $\hat{N}_2 = 0$ and $\hat{N}_2 = 400$. In the first case we then have an $\hat{N}_1$ value of $\hat{N}_1 = 1000 - 2 \cdot 0 = 1000$, and in the second case we have an $\hat{N}_1$ value of $\hat{N}_1 = 1000 - 2 \cdot 400 = 200$. Therefore a third equilibrium is (iii) $\hat{N}_1 = 1000$, $\hat{N}_2 = 0$ and a fourth is (iv) $\hat{N}_1 = 200$, $\hat{N}_2 = 400$ as shown in Figure 3(c).

(c) With $\beta = 0$, the equilibria are now pairs of values, $(\hat{N}_1, \hat{N}_2)$ that simultaneously satisfy the equations

$$\left(1 - \frac{\hat{N}_1 + 2\hat{N}_2}{1000}\right)\hat{N}_1 = 0 \quad \text{and} \quad \left(1 - \frac{\hat{N}_2}{600}\right)\hat{N}_2 = 0$$

The second equation no longer involves $\hat{N}_1$ and therefore we can solve it immediately for $\hat{N}_2$. We obtain $\hat{N}_2 = 0$ and $\hat{N}_2 = 600$.

Substituting $\hat{N}_2 = 0$ into the first equation gives $[1 - \hat{N}_1/1000]\hat{N}_1 = 0$. Solving this for $\hat{N}_1$ gives $\hat{N}_1 = 0$ and $\hat{N}_1 = 1000$. Therefore two equilibria are (i) $\hat{N}_1 = 0$, $\hat{N}_2 = 0$ and (ii) $\hat{N}_1 = 1000$, $\hat{N}_2 = 0$.

If instead we substitute $\hat{N}_2 = 600$ into the first equation, we get

$$\left(1 - \frac{\hat{N}_1 + 1200}{1000}\right)\hat{N}_1 = 0$$

Solving this for $\hat{N}_1$ gives $\hat{N}_1 = 0$ and $\hat{N}_1 = -200$. Therefore two additional equilibria are (iii) $\hat{N}_1 = 0$, $\hat{N}_2 = 600$ and (iv) $\hat{N}_1 = -200$, $\hat{N}_2 = 600$.

Notice that equilibrium (iv) involves a negative value of $\hat{N}_1$. From a mathematical standpoint this is a perfectly fine equilibrium, but from a biological standpoint it is not of interest because it would correspond to a negative population size. Equilibria that are biologically relevant [(i), (ii), and (iii) in this example] are referred to as **biologically feasible**. ■

■ Qualitative Dynamics in the Phase Plane

Let's return to Figure 2 from the predator-prey model of Equations 1. In Figure 2(a) we plotted the R -nullclines. These are curves in the plane along which $dR/dt = 0$.

Therefore these curves separate the plane into regions within which either $dR/dt > 0$ or $dR/dt < 0$. We can determine which of these two situations applies in each region. Consider the region above the line $W = r/a$ in Figure 2(a). This corresponds to large values of W , in which case Equations 1 give

$$\frac{dR}{dt} = rR - aRW \approx -aRW < 0$$

for large enough values of W . Therefore $dR/dt < 0$ in this region. Conversely, if W is close to zero, we have

$$\frac{dR}{dt} = rR - aRW \approx rR > 0$$

We can therefore indicate whether R is increasing or decreasing in each of these regions of the phase plane with a single arrow as in Figure 4(a).

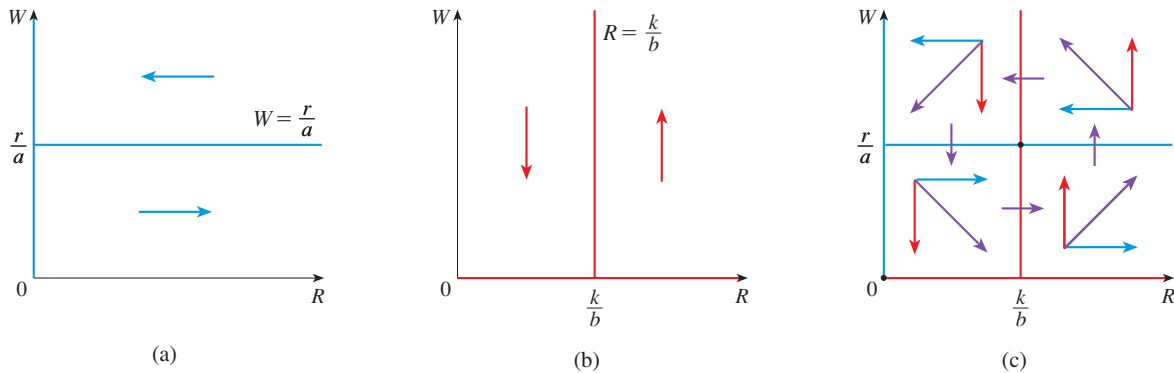


FIGURE 4
Purple arrows indicate direction of motion in part (c).

We can follow the same procedure for the W -nullclines in Figure 2(b). To the right of the line $R = k/b$ the value of R will be very large. From Equations 1 we have

$$dW/dt = -kW + bRW \approx bRW > 0$$

Conversely, if R is close to zero then

$$dW/dt = -kW + bRW \approx -kW < 0$$

This gives the direction arrows for W in Figure 4(b). Putting these two plots together then gives the overall direction of movement by the purple arrows in the phase plane shown in Figure 4(c). This provides a very general, qualitative picture of the dynamics without having to plot a large number of direction arrows. In this case we see that spiraling trajectories in the phase plane are expected.

EXAMPLE 2 | Lotka-Volterra competition equations (continued)

- Determine the qualitative dynamics in the phase plane for the Lotka-Volterra competition equations of Example 1.
- Plot the variables as a function of time.

SOLUTION

- We begin by first considering the N_1 -nullclines shown in Figure 3(a). Above the nullcline the value of N_2 will be very large. From Equations 3 we have

$$\frac{dN_1}{dt} = \left(1 - \frac{N_1 + 2N_2}{1000}\right)N_1 \approx -\frac{2N_2}{1000}N_1 < 0$$

for large enough N_2 . But as we move closer to the origin, N_1 and N_2 will be very small. From Equations 3 we have

$$\frac{dN_1}{dt} = \left(1 - \frac{N_1 + 2N_2}{1000}\right)N_1 \approx N_1 > 0$$

[See Figure 5(a).]

For the N_2 -nullclines, a similar argument shows that to the right of the N_2 -nullcline, $dN_2/dt < 0$. Likewise, as we move close to the origin, $dN_2/dt > 0$ [see Figure 5(b)]. Putting together these two plots gives the qualitative dynamics shown in Figure 5(c). This reveals that equilibria (ii) and (iii) are both locally stable (if we start near either of these equilibria, we will move toward the equilibrium). And we can see that equilibria (i) and (iv) are unstable (if we start near either of these, we will move away). Thus, the two species do not coexist. One will competitively exclude the other, and the initial conditions determine which species “wins.” Exercise 27 explores this model for other parameter values.

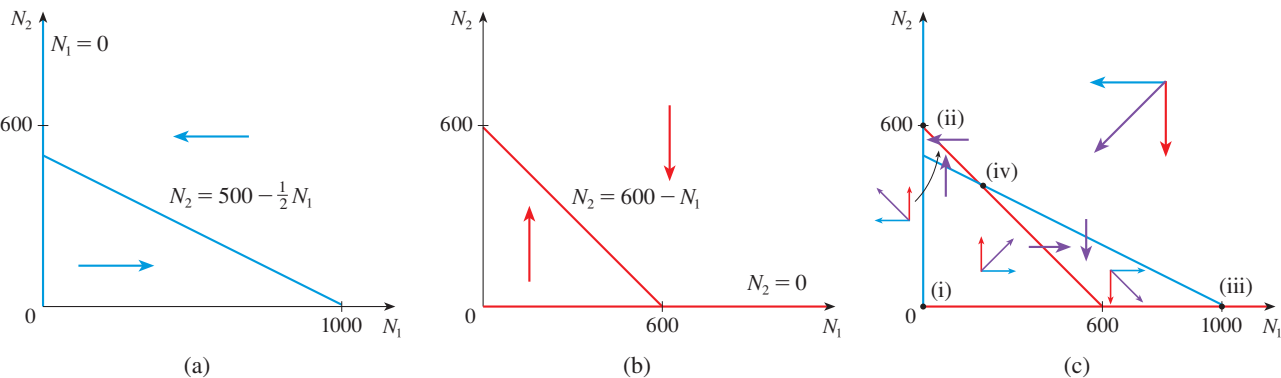


FIGURE 5
Purple arrows indicate direction of motion in part (c).

(b) In Figure 6 we have used a CAS to plot the variables against time for two sets of initial conditions. In part (a) N_2 initially increases but then decays to zero while N_1 continually increases. In part (b) the opposite occurs. These correspond to the variables moving toward equilibria (iii) and (ii) in Figure 5(c), respectively.

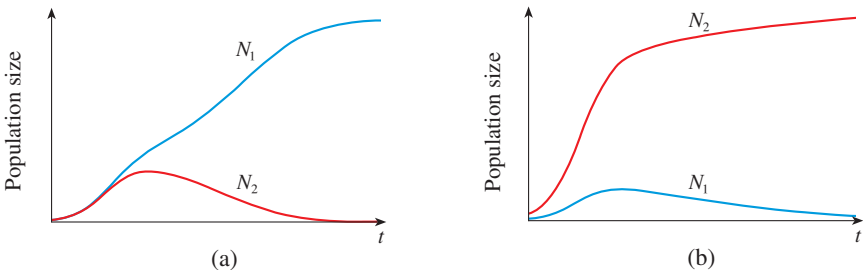


FIGURE 6

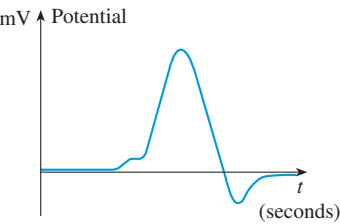


FIGURE 7
Action potential

EXAMPLE 3 | Fitzhugh-Nagumo equations for a neuron Neurons carry electrical impulses throughout the body and display what is called an *all-or-nothing* response. Low levels of electrical stimulation have little effect on the neuron. If the electrical stimulus rises above a certain threshold intensity, however, a large electrical impulse called an action potential is generated and travels along the length of the neuron (see Figure 7). Increasing the level of electrical stimulation further does not seem to alter the impulse.

The electrical potential of a neuron has a resting value (which we take to be zero) and a threshold value a above which an impulse is triggered. When an impulse

is triggered it grows in potential to a maximum possible value that we take to be 1. Using $v(t)$ for the neuron's potential (in mV) at time t (in seconds), its initial dynamics are therefore similar to those of population dynamics with an Allee effect (recall Example 7.2.3); namely, $dv/dt = (v - a)(1 - v)v$, where a is a constant satisfying $0 < a < 1$. The potential eventually returns to its resting state, however, as the permeability of the neuron's cell wall changes and allows exchange of charged molecules (ions of potassium and sodium). This behavior can be modeled by simply appending a loss term to the equation for the dynamics of v . Assuming the rate of loss is proportional to the magnitude of ion exchange, and using w to denote this magnitude of ion exchange, the system is therefore

$$(4) \quad \frac{dv}{dt} = (v - a)(1 - v)v - w \quad \frac{dw}{dt} = bv - cw$$

In the 1950s Alan Lloyd Hodgkin and Andrew Fielding Huxley developed a model for neuron activity based on a system of four differential equations. This work, coupled with their study of squid neurons, earned them the 1963 Nobel Prize in Medicine and Physiology. The Fitzhugh-Nagumo equations are a simplified, two-dimensional version of their model.

Both v and w can be positive or negative. Notice that the dampening effect of ion exchange w increases in proportion to the potential at rate b , and it is also subject to a constant per unit decay rate of c .

- Suppose $a = 0.2$, $b = 0.01$, and $c = 0.04$. Identify the equilibria in the phase plane and determine the qualitative dynamics.
- Plot the potential $v(t)$ as a function of time.

SOLUTION

(a) The v -nullclines satisfy $dv/dt = 0$, that is, $w = (v - 0.2)(1 - v)v$, a cubic equation whose graph crosses the horizontal axis at $v = 0$, $v = 0.2$, and $v = 1$ [see Figure 8(a)]. Rather than substituting extreme values of the variables as we did in Examples 1 and 2, here we use an alternative method to determine the direction of movement in the phase plane.

From Equations 4 we can see that $dv/dt > 0$ whenever $w < (v - 0.2)(1 - v)v$. Since $(v - 0.2)(1 - v)v$ defines the cubic polynomial in Figure 8(a), v will therefore be increasing below the curve and decreasing above it.

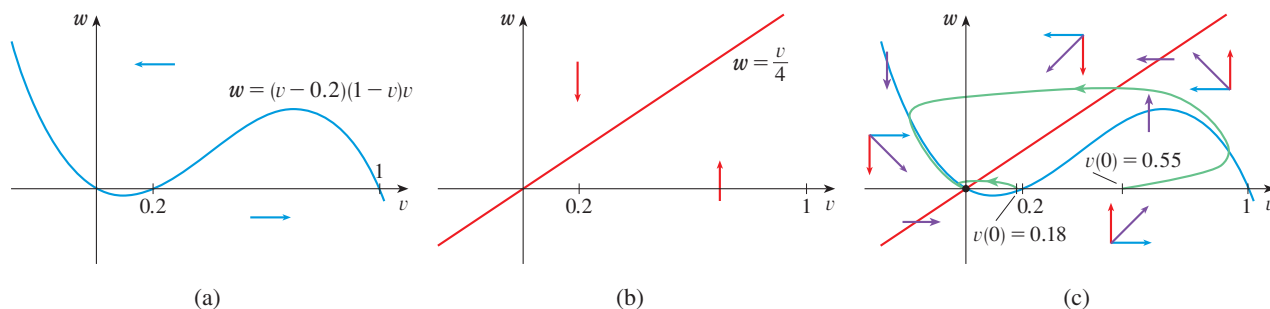


FIGURE 8
Purple arrows indicate direction of motion. Green curves are solution curves.

The w -nullcline satisfies $0.01v - 0.04w = 0$, or $w = v/4$ as in Figure 8(b). From Equations 4 we can see that $dw/dt > 0$ whenever $0.01v > 0.04w$. We can rewrite this inequality as $w < v/4$, and since $w = v/4$ defines the line plotted in Figure 8(b), w will be increasing below the line and decreasing above it.

The nullclines are plotted together, along with the single equilibrium at the origin, in Figure 8(c): this shows that the system exhibits an oscillatory behavior around the equilibrium. But the qualitative nature of these oscillatory dynamics depends on the initial conditions. Using a CAS, we have plotted two solution curves, both with $w(0) = 0$. If the initial potential $v(0)$ is below the threshold of 0.2, the neuron potential immediately decays towards zero. If the initial potential is above 0.2, then the potential initially grows in magnitude before eventually decaying back to the equilibrium state.

(b) The potential for each set of initial conditions in Figure 8(c) is plotted against time in Figure 9 using a CAS. Figure 9(b) resembles an actual action potential when the initial stimulus of the neuron is above the threshold value.

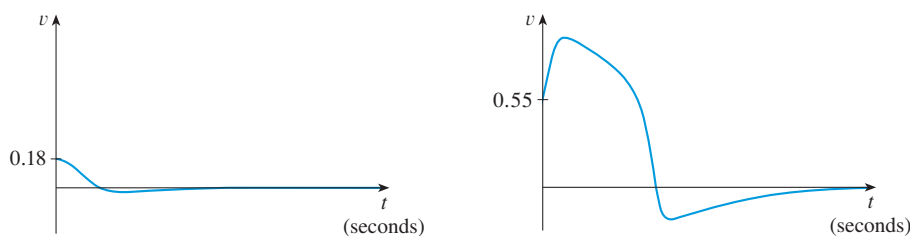


FIGURE 9

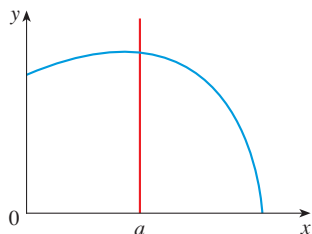
(a) $v(0) = 0.18, w(0) = 0$ (b) $v(0) = 0.55, w(0) = 0$

As a final remark, we note that a phase plane analysis can sometimes give us information about the stability of equilibria. In Example 2 the analysis gave us conclusive information about the stability properties of all the equilibria. In Example 3, however, the phase plane analysis itself is insufficient to allow us to reach any conclusion (although the equilibrium is, in fact, locally stable). We can determine the qualitative tendency to oscillate around the equilibrium point, but we cannot determine whether these oscillations converge towards the equilibrium or move away from it. The same is true for the predator-prey model in Figure 4. In Section 10.4 we will derive mathematical criteria that distinguish between these possibilities.

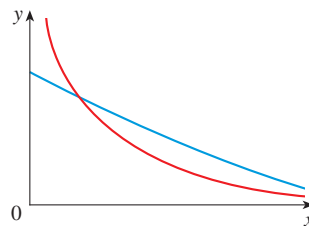
EXERCISES 7.6

1–6 In each phase plane the x -nullclines are blue and the y -nullclines are red. Use the information given to indicate the direction of movement in the phase plane and label all equilibria. For each equilibrium, determine if it is locally stable or unstable, or if the information is inconclusive.

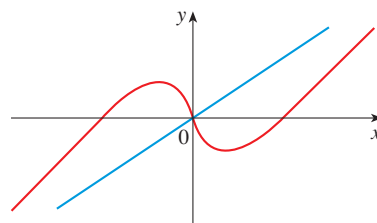
- The variable x is increasing in the region below the curved nullcline and decreasing elsewhere. The variable y is decreasing between $x = 0$ and $x = a$ and increasing elsewhere.



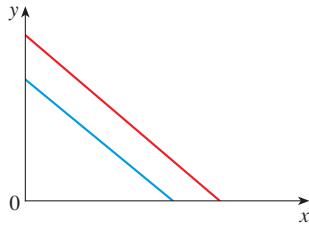
- The variable x is increasing between zero and its nullcline and decreasing elsewhere. The variable y is increasing below its nullcline and decreasing above it.



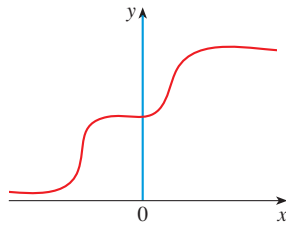
- The variable x is increasing above its nullcline and decreasing below it. The variable y is decreasing above its nullcline and increasing below it.



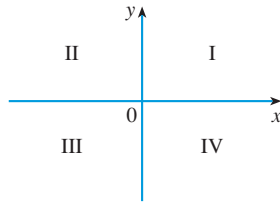
- The variable x is increasing below its nullcline and decreasing above it. The variable y is decreasing above its nullcline and increasing below it.



5. The variable x is increasing to the left of its nullcline and decreasing to the right of it. The variable y is decreasing above its nullcline and increasing below it.



6. The variable x is increasing in quadrants I and III and decreasing in quadrants II and IV. The variable y is increasing everywhere.



7–15 A system of differential equations is given.

- (a) Construct the phase plane, plotting all nullclines, labeling all equilibria, and indicating the direction of motion.
(b) Obtain an expression for each equilibrium.

7. $x' = x(3 - x - y)$, $y' = y(2 - x - y)$, $x, y \geq 0$
8. $p' = p(1 - p - q)$, $q' = q(2 - 3p - q)$, $p, q \geq 0$
9. $n' = n(1 - 2m)$, $m' = m(2 - 2n - m)$, $n, m \geq 0$
10. $x' = x(2 - x)$, $y' = y(3 - y)$
11. $p' = -p^2 + q - 1$, $q' = q(2 - p - q)$
12. $p' = 2q - 1$, $q' = q^2 - q - p$
13. $x' = 5 - 2x - xy$, $y' = xy - y$, $x, y \geq 0$
14. $z' = z^3 - 4z^2 + 3z - 2w$, $w' = z - w - 1$
15. $x' = -(x - 2) \ln(xy)$, $y' = e^x(x - y)$, $x, y > 0$

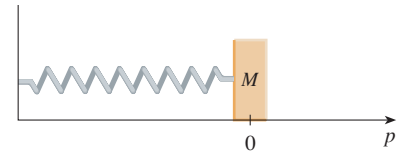
16–20 A system of differential equations is given.

- (a) Use a phase plane analysis to determine the values of the constant a for which the sole equilibrium of the differential equations is locally stable.

- (b) Obtain an expression for each equilibrium (it may be a function of the constant a).

16. $x' = a(x - 3)$, $y' = 5 - y$, $a \neq 0$
17. $x' = y - ax$, $y' = x - y$, $a > 0$, $a \neq 1$
18. $x' = a(x - a)$, $y' = 4 - y - x$, $a \neq 0$
19. $x' = ay^2 - x + 1$, $y' = 2(1 - y)$
20. $x' = -(y - 1) - a(x - 1)$,
 $y' = -(y - 1) - \frac{1}{a}(x - 1)$, $a \neq -1, 0, 1$, $x, y > 0$

21. Hooke's Law states that the force F exerted by a spring on a mass is proportional to the displacement from its resting position.



From the figure we have $F = -kp$ for some positive constant k , where p is position. Also, Newton's Second Law tells us that $F = ma$, where m is the mass of the object and a is its acceleration. Further, since we know that acceleration is the second derivative of position with respect to time, we obtain the differential equation

$$m \frac{d^2 p}{dt^2} = -kp$$

This is a **second-order differential equation** because it involves the second derivative of the unknown function, p .

- (a) Define a new variable $q = dp/dt$. Show that the second-order differential equation can then be written as the following system of two first-order differential equations

$$\frac{dp}{dt} = q \quad \frac{dq}{dt} = -\frac{kp}{m}$$

- (b) Construct the phase plane for the equations in part (a), including nullclines, the equilibrium, and the direction of movement.
(c) What does the phase plane analysis from part (b) tell you about the position of the mass over time?
(d) What does the phase plane analysis from part (b) tell you about the velocity of the mass over time?
22. The **van der Pol equation** is a second-order differential equation describing oscillatory dynamics in a variable x :

$$\frac{dx^2}{dt^2} - \mu(1 - x^2) \frac{dx}{dt} + x = 0$$

where μ is a positive constant. This equation was first obtained by an electrical engineer named Balthasar van der Pol, but has since been used as a model for a

variety of phenomena, including sustained oscillatory dynamics of neural impulses.

- (a) Convert the Van der Pol equation into a system of two first-order differential equations by defining the new variable y as

$$y = x - \frac{x^3}{3} - \frac{dx/dt}{\mu}$$

- (b) Construct the phase plane for the equations obtained in part (a), including nullclines, the equilibrium, and the direction of movement.
- (c) What does the phase plane analysis from part (b) tell you about the dynamics of x ?

- 23. The Kermack-McKendrick equations** are first-order differential equations describing an infectious disease outbreak. Using S and I to denote the number of susceptible and infected people in a population, the equations are

$$S' = -\beta SI \quad I' = \beta SI - \mu I$$

where β and μ are positive constants representing the transmission rate and rate of recovery, respectively.

- (a) Provide a biological explanation for each term of the equations.
- (b) Suppose $\beta = 1$ and $\mu = 5$. Construct the phase plane including all nullclines, equilibria, and arrows indicating the direction of movement in the plane.
- (c) Construct the phase plane for arbitrary values of β and μ , including all nullclines, equilibria, and arrows indicating direction of movement in the plane.

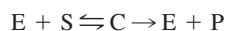
- 24. The Kermack-McKendrick equations** from Exercise 23 can be extended to model persistent diseases rather than single outbreaks by including an inflow of susceptible individuals and their natural death. This gives the differential equations

$$S' = \theta - \gamma S - \beta SI \quad I' = \beta SI - \mu I$$

where θ and γ are positive constants representing inflow and mortality of susceptible individuals, respectively.

- (a) Suppose $\beta = 1$, $\mu = 1$, $\gamma = 1$, and $\theta = 10$. Construct the phase plane including all nullclines, equilibria, and arrows indicating the direction of movement in the plane.
- (b) Suppose $\beta = \frac{1}{15}$, $\mu = 1$, $\gamma = 1$, and $\theta = 10$. Construct the phase plane including all nullclines, equilibria, and arrows indicating the direction of movement in the plane.
- (c) What is the difference in the predicted dynamics between part (a) and part (b)?

- 25. The Michaelis-Menten equations** describe a biochemical reaction in which an enzyme E and substrate S bind to form a complex C . This complex can then either dissociate back into its original components or undergo a reaction in which a product P is produced along with the free enzyme:



This can be expressed by the differential equations

$$\frac{dx}{dt} = -k_f xyM + k_r(1 - y)M$$

$$\frac{dy}{dt} = -k_f xyM + k_r(1 - y)M + k_{cat}(1 - y)M$$

$$\frac{dz}{dt} = k_{cat}(1 - y)M$$

where M is the total number of enzymes (both free and bound), x and z are the numbers of substrate and product molecules, y is the fraction of the enzyme pool that is free, and the k_i 's are positive constants.

- (a) Explain all the terms in this system of differential equations.
- (b) Although this is a system of three differential equations, its dynamics can be understood by constructing a phase plane for the variables x and y alone. Explain why.
- (c) Construct the phase plane mentioned in part (b), including all nullclines and equilibria, and indicate the direction of movement in the plane.

- 26. Metastasis of malignant tumors** Metastasis is the process by which cancer cells spread throughout the body and initiate tumors in various organs. This sometimes happens via the bloodstream, where cancer cells become lodged in capillaries of organs and then move across the capillary wall into the organ. Using C to denote the number of cells lodged in a capillary and I for the number that have invaded the organ, we can model this as

$$C' = -\alpha C - \beta C \quad I' = \alpha C - \delta I + \rho I$$

where all constants are positive, α is the rate of movement across the capillary wall, β is the rate of dislodgment from the capillary, δ is the rate at which cancer cells in the organ die, and ρ is their growth rate.

- (a) Suppose $\rho < \delta$. Construct the phase plane, including all nullclines, equilibria, and arrows indicating the direction of movement in the plane.
- (b) Suppose $\rho > \delta$. Construct the phase plane, including all nullclines, equilibria, and arrows indicating the direction of movement in the plane.
- (c) What is the difference in the predicted dynamics between part (a) and part (b)?

Source: Adapted from D. Kaplan et al., *Understanding Nonlinear Dynamics* (New York: Springer-Verlag, 1995).

- 27. Lotka-Volterra competition equations** For each case, derive the equations for all nullclines of the Lotka-Volterra model in Example 1 and use them to construct the phase plane, including all nullclines, equilibria, and arrows indicating the direction of movement. (Assume that all constants are positive.)

- (a) $K_1 > \alpha K_2$ and $K_2 < \beta K_1$
- (b) $K_1 < \alpha K_2$ and $K_2 > \beta K_1$
- (c) $K_1 < \alpha K_2$ and $K_2 < \beta K_1$
- (d) $K_1 > \alpha K_2$ and $K_2 > \beta K_1$

28–30 Consumer-resource models often have the following general form

$$R' = f(R) - g(R, C) \quad C' = \varepsilon g(R, C) - h(C)$$

where $f(R)$ is a function describing the rate of replenishment of the resource, $g(R, C)$ describes the rate of consumption of the resource, and $h(C)$ is the rate of loss of the consumer. The constant ε is the conversion efficiency of resources into consumers and lies between zero and one. Construct the phase plane, including all nullclines, equilibria, and arrows indicating the direction of movement in the plane. Describe how consumer and resource abundance are predicted to change over time.

28. A chemostat is an experimental consumer-resource system.

If the resource is not self-reproducing, then it can be modeled by choosing $f(R) = \theta$, $g(R, C) = bRC$, and $h(C) = \mu C$, where all constants are positive.

29. A model for self-reproducing resources is obtained by choosing $f(R) = rR$, $g(R, C) = bRC$, and $h(C) = \mu C$, where all constants are positive.

30. A model for self-reproducing resources with limited growth is obtained by choosing $f(R) = rR(1 - R/K)$, $g(R, C) = bRC$, and $h(C) = \mu C$. Assume all constants are positive and $K > \mu/(\varepsilon b)$.

31. Hemodialysis is a process by which a machine is used to filter urea and other waste products from a patient's blood if their kidneys fail. The concentration of a patient's urea during dialysis is sometimes modeled by supposing there are two compartments within the patient—the blood, which is directly filtered by the dialysis machine, and another compartment that cannot be directly filtered but that is connected to the blood. A system of two differential

equations describing this is

$$\frac{dc}{dt} = -\frac{K}{V}c + ap - bc \quad \frac{dp}{dt} = -ap + bc$$

where c and p are the urea concentrations in the blood (in mg/mL) and in the inaccessible pool, respectively, and all constants are positive.

- Explain each term of the system of differential equations.
- Construct the phase plane, including all nullclines, equilibria, and arrows indicating the direction of movement in the plane. What happens to the urea concentration as $t \rightarrow \infty$?

32. Fitzhugh-Nagumo equations Consider the following alternative form of the Fitzhugh-Nagumo equations from Example 3:

$$\frac{dv}{dt} = (v - a)(1 - v)v - w \quad \frac{dw}{dt} = \varepsilon(v - w)$$

where $\varepsilon > 0$ and $0 < a < 1$. Construct the phase plane, including all nullclines, equilibria, and arrows indicating the direction of movement in the plane.

33. The Rosenzweig-MacArthur model is a consumer-resource model similar to that from Exercise 30, but with a different consumption function. A simplified version is

$$R' = R(K - R) - \frac{R}{a + R}C \quad C' = \frac{R}{a + R}C - bC$$

Suppose that all constants are positive and that $K > ab/(1 - b) > 0$. Construct the phase plane, including all nullclines, equilibria, and arrows indicating the direction of movement in the plane.

PROJECT Determining the Critical Vaccination Coverage



© Tatik22 / Shutterstock.com

Vaccines are preventative medications that are administered before an individual becomes infected by a pathogen. Either vaccinated people do not become infected or they are able to clear the infection more quickly. Let's suppose a vaccine shortens the duration of the infection by causing the vaccinated individual's immune system to clear the infection more quickly. We will model the dynamics of infected people only, using the following pair of autonomous differential equations:

$$\begin{aligned} \frac{dN}{dt} &= N\beta S(1 - p) + V\beta S(1 - p) - (m + c)N \\ \frac{dV}{dt} &= N\beta Sp + V\beta Sp - (m + c\gamma)V \end{aligned}$$

where N and V are the numbers of nonvaccinated and vaccinated individuals, respectively, S is the number of susceptible individuals, and p is the fraction of these that are vaccinated (with $1 - p$ remaining unvaccinated). For simplicity, these equations assume that the total number of susceptible people remains approximately constant at S over the time frame of interest, and that all infected individuals suffer a constant per capita

mortality rate of m regardless of vaccination status. The constant β quantifies the transmissibility of the disease and is positive. Infected individuals are removed through their clearing of the infection via an immune response, with per capita rate c for unvaccinated individuals and $c\gamma$ for vaccinated individuals (with $\gamma > 1$, which reflects the heightened clearance rate caused by the vaccine).

Our task is to determine the value of p needed to ensure that the epidemic will die out, written as a function of the vaccine effectiveness γ .

1. Explain all the terms in the differential equations.
2. Let's assume that $\beta S > m + c$. What does this imply about the disease dynamics if nobody is vaccinated (that is, if $p = 0$)? Let's also assume that $\beta S < m + c\gamma$. What does this imply about the disease dynamics if everybody is vaccinated (that is, if $p = 1$)?
3. Assuming $\beta S > m + c$ and $\beta S < m + c\gamma$, construct the phase plane and nullclines for the case where the vaccination coverage is very low (that is, p is very small). Indicate the sole equilibrium.
4. Determine whether the equilibrium found in Problem 3 is stable or not using the phase plane.
5. As you increase the vaccination coverage p from near zero to near one, the nullclines move. Determine the direction in which each nullcline moves as p increases.
6. From the phase plane in Problem 3 and your answer to Problem 5 you should be able to determine that there is a critical value of p greater than which the number of infected individuals always decreases to zero. Illustrate a phase plane diagram for the case where p is just less than this critical value, and another for the case where p is just greater than this critical value.
7. Use your result from Problem 6 to obtain a mathematical inequality that must be satisfied by p for the vaccination coverage to be adequate to prevent the epidemic. Your inequality should involve the vaccine effectiveness γ .

Chapter 7 REVIEW

CONCEPT CHECK

1. (a) What is a differential equation?
(b) What is the order of a differential equation?
(c) What is an initial condition?
(d) What are the differences between pure-time, autonomous, and nonautonomous differential equations?
2. What can you say about the solutions of the equation $y' = x^2 + y^2$ just by looking at the differential equation?
3. What is a phase plot for the differential equation $y' = g(y)$?
4. What is a direction field for the differential equation $y' = F(x, y)$?
5. Explain how Euler's method works.
6. What is a separable differential equation? How do you solve it?
7. (a) Write the logistic equation.
(b) Under what circumstances is this an appropriate model for population growth?
8. (a) Write Lotka-Volterra equations to model populations of sharks S and their food F .
(b) What do these equations say about each population in the absence of the other?
9. What is a nullcline?
10. (a) Write Lotka-Volterra competition equations for two competing fish species, x and y .
(b) What would the nullclines have to look like for species x to always outcompete species y ?

Answers to the Concept Check can be found on the back endpapers.

TRUE-FALSE QUIZ

Determine whether the statement is true or false. If it is true, explain why. If it is false, explain why or give an example that disproves the statement.

1. All solutions of the differential equation $y' = -1 - y^4$ are decreasing functions.
2. The function $f(x) = (\ln x)/x$ is a solution of the differential equation $x^2 y' + xy = 1$.
3. Consider the differential equation $y' = g(y)$ where $g(y)$ is a differentiable function of y . It is not possible for y to exhibit oscillatory behavior.

4. The equation $y' = x + y$ is separable.
5. The equation $y' = 3y - 2x + 6xy - 1$ is separable.
6. If y is the solution of the initial-value problem

$$\frac{dy}{dt} = 2y \left(1 - \frac{y}{5} \right) \quad y(0) = 1$$

then $\lim_{t \rightarrow \infty} y = 5$.

EXERCISES

1–4 A differential equation is given.

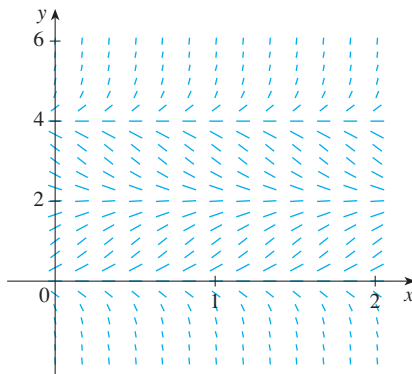
- (a) Determine all equilibria as a function of the constant a .
- (b) Construct a phase plot and use it to determine the stability of the equilibria found in part (a) for three different values of the constant: (i) $a < 0$, (ii) $a = 0$, and (iii) $a > 0$.
- (c) Use the local stability criterion to verify your answers to part (b).

- | | |
|--------------------|--------------------|
| 1. $x' = ax - x^2$ | 2. $x' = a - x^2$ |
| 3. $x' = ax - x^3$ | 4. $x' = ax + x^3$ |

5. (a) A direction field for the differential equation $y' = y(y - 2)(y - 4)$ is shown. Sketch the graphs of the solutions that satisfy the given initial conditions.

(i) $y(0) = -0.3$	(ii) $y(0) = 1$
(iii) $y(0) = 3$	(iv) $y(0) = 4.3$

 (b) If the initial condition is $y(0) = c$, for what values of c is $\lim_{t \rightarrow \infty} y(t)$ finite? What are the equilibrium solutions?

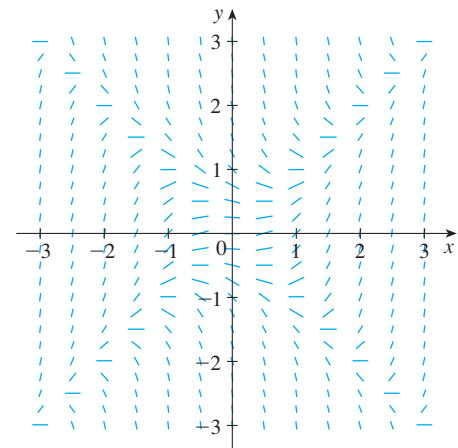


6. (a) Sketch a direction field for the differential equation $y' = x/y$. Then use it to sketch the four solutions that satisfy the initial conditions $y(0) = 1$, $y(0) = -1$, $y(2) = 1$, and $y(-2) = 1$.

- (b) Check your work in part (a) by solving the differential equation explicitly. What type of curve is each solution curve?
7. (a) A direction field for the differential equation $y' = x^2 - y^2$ is shown. Sketch the solution of the initial-value problem

$$y' = x^2 - y^2 \quad y(0) = 1$$

Use your graph to estimate the value of $y(0.3)$.



- (b) Use Euler's method with step size 0.1 to estimate $y(0.3)$, where $y(x)$ is the solution of the initial-value problem in part (a). Compare with your estimate from part (a).
 - (c) On what lines are the centers of the horizontal line segments of the direction field in part (a) located? What happens when a solution curve crosses these lines?
8. (a) Use Euler's method with step size 0.2 to estimate $y(0.4)$, where $y(x)$ is the solution of the initial-value problem

$$y' = 2xy^2 \quad y(0) = 1$$
 (b) Repeat part (a) with step size 0.1.

- (c) Find the exact solution of the differential equation and compare the value at 0.4 with the approximations in parts (a) and (b).

9–10 Solve the differential equation.

$$9. 2ye^{y^2}y' = 2x + 3\sqrt{x} \quad 10. \frac{dx}{dt} = 1 - t + x - tx$$

11–12 Solve the initial-value problem.

$$11. \frac{dr}{dt} + 2tr = r, \quad r(0) = 5$$

$$12. (1 + \cos x)y' = (1 + e^{-y})\sin x, \quad y(0) = 0$$

- 13. Seasonality and population dynamics** The per capita growth rate of a population varies seasonally. The population dynamics are modeled as

$$n' = \cos\left(\frac{2\pi t}{365}\right)n \quad n(0) = n_0$$

where $n(t)$ is the population size at time t (measured in days). Determine the population size at time t .

- 14. Seasonality and population dynamics** The per capita growth rate of a population varies seasonally and habitat destruction is also occurring. This is modeled as

$$n' = r\left(\cos\left[\frac{2\pi t}{365}\right] - at\right)n \quad n(0) = n_0$$

where $n(t)$ is the population size at time t (measured in days) and r and a are positive constants. Determine the population size at time t .

- 15. Levins' metapopulation model** from Exercise 7.2.15 describes a population consisting of patches that can be either occupied or vacant. Occupied patches create more occupied patches by sending individuals to unoccupied patches. If the frequency of occupied patches is p , we would therefore expect that the rate at which new patches become occupied is proportional to both p and $1 - p$. Patches also become unoccupied through mortality at a constant rate. The differential equation for p is

$$\frac{dp}{dt} = cp(1 - p) - mp \quad p(0) = p_0$$

where c and m are constants.

- (a) Find the solution to this initial-value problem.
(b) Under what conditions on the constants will the frequency of occupied patches go to zero as $t \rightarrow \infty$?

- 16. The Brentano-Stevens Law** in psychology models the way that a subject reacts to a stimulus. It states that if R repre-

sents the reaction to an amount S of stimulus, then the relative rates of increase are proportional:

$$\frac{1}{R} \frac{dR}{dt} = \frac{k}{S} \frac{dS}{dt}$$

where k is a positive constant. Find R as a function of S .

- 17. Lung preoxygenation** Some medical procedures require a patient's airway to be temporarily blocked, preventing the inspiration of oxygen. The duration of time over which such procedures can be performed safely may be increased by replacing a large fraction of the air in the patient's lungs with oxygen prior to the procedure. Suppose the lung volume is 3 L and well-mixed air in the lungs is replaced with pure oxygen at a rate of 10 mL/s.

- (a) What is the amount of oxygen in the lungs as a function of time if they initially contain 20% oxygen?
(b) Use your answer from part (a) to determine how long the process of oxygenation should be run to result in an 80% oxygen content in the lungs.

- 18.** A tank contains 100 L of pure water. Brine that contains 0.1 kg of salt per liter enters the tank at a rate of 10 L/min. The solution is kept thoroughly mixed and drains from the tank at the same rate. How much salt is in the tank after six minutes?

- 19. Hormone transport** In lung physiology, the transport of a substance across a capillary wall has been modeled by the differential equation

$$\frac{dh}{dt} = -\frac{R}{V} \left(\frac{h}{k + h} \right)$$

where h is the hormone concentration in the bloodstream (in mg/mL), t is time (in seconds), R is the maximum transport rate, V is the volume of the capillary, and k is a positive constant that measures the affinity between the hormones and the enzymes that assist the process. Solve this differential equation to find a relationship between h and t .

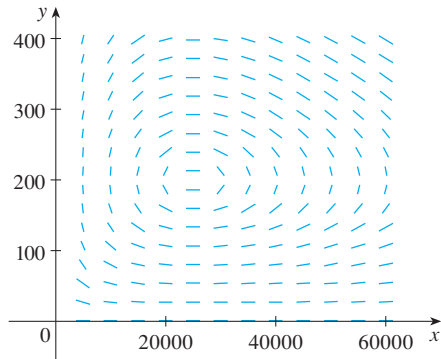
- 20. Predator-prey dynamics** Populations of birds and insects are modeled by the equations

$$\frac{dx}{dt} = 0.4x - 0.002xy$$

$$\frac{dy}{dt} = -0.2y + 0.000008xy$$

- (a) Which of the variables, x or y , represents the bird population, and which represents the insect population? Explain.
(b) Find the equilibrium solutions and explain their significance.
(c) Find an expression for dy/dx .
(d) The direction field for the differential equation in part (c) is shown. Use it to sketch the phase trajectory corre-

sponding to initial populations of 100 birds and 40,000 insects. Then use the phase trajectory to describe how both populations change.



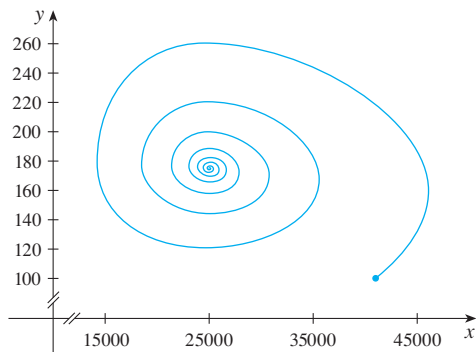
- (e) Use part (d) to make rough sketches of the bird and insect populations as functions of time. How are these two graphs related?

21. Suppose the model of Exercise 20 is replaced by the equations

$$\frac{dx}{dt} = 0.4x(1 - 0.000005x) - 0.002xy$$

$$\frac{dy}{dt} = -0.2y + 0.000008xy$$

- (a) According to these equations, what happens to the insect population in the absence of birds?
 (b) Find the equilibrium solutions and explain their significance.
 (c) The figure shows the phase trajectory that starts with 100 birds and 40,000 insects. Describe what eventually happens to the bird and insect populations.



- (d) Sketch graphs of the bird and insect populations as functions of time.

22. **Cancer progression** The development of many cancers, such as colorectal cancer, proceeds through a series of

precancerous stages. Suppose there are $n - 1$ precancerous stages before cancer develops at stage n . A simple system of differential equations modeling this is

$$x'_0 = -u_0x_0$$

$$x'_i = u_{i-1}x_{i-1} - u_ix_i$$

$$x'_n = u_{n-1}x_{n-1}$$

where x_i is the fraction of the population in state i , the u_i 's are constants, and $i = 1, \dots, n - 1$.

- (a) Suppose $n = 2$. What is the system of three differential equations?
 (b) Suppose $u_0 = 1$ and $u_1 = 1$. Construct the phase plane for the variables x_0 and x_1 alone, including all nullclines, equilibria, and arrows indicating the direction of movement in the plane.
 (c) From your answer to part (a), obtain a differential equation for dx_1/dx_0 for the parametric curves describing the trajectories in the x_0 - x_1 plane. Use this equation to plot the vector field. Then sketch several solutions curves for x_1 as a function of x_0 , assuming that $x_0(0) = a$ and $x_1(0) = 1 - a$, where $0 < a < 1$.
 (d) Your answer to part (c) will produce a family of functions in the constant a . Provide a biological interpretation for these parametric curves.

23. **Competition-colonization models** The metapopulation model from Exercise 15 can be extended to include two species, where one is a superior competitor. The equations are

$$\frac{dp_1}{dt} = c_1p_1(1 - p_1) - m_1p_1$$

$$\frac{dp_2}{dt} = c_2p_2(1 - p_1 - p_2) - m_2p_2 - c_1p_1p_2$$

where p_1 and p_2 are the fractions of patches occupied by species 1 and 2, respectively. These equations model a process in which any patch has at most one species, and where species 2 patches can be “taken over” by species 1, but not vice versa.

- (a) Explain how the terms in the equations reflect the assumption that species 1 is the superior competitor.
 (b) Suppose that $m_1 = m_2 = 3$, $c_1 = 5$, and $c_2 = 30$. Construct the phase plane, including all nullclines, equilibria, and arrows indicating the direction of movement in the plane. Show that, despite species 1 being a better competitor, the two species are predicted to coexist.

24. **Habitat destruction** The model of Exercise 23 can be extended to include the effects of habitat destruction. Suppose that only a fraction h of the patches are habitable

($0 < h < 1$). The equations become

$$\frac{dp_1}{dt} = c_1p_1(h - p_1) - m_1p_1$$

$$\frac{dp_2}{dt} = c_2p_2(h - p_1 - p_2) - m_2p_2 - c_1p_1p_2$$

Suppose that $m_1 = m_2 = 3$, $c_1 = 5$, and $c_2 = 30$.

- (a) Construct the phase plane, including all nullclines, equilibria, and arrows indicating the direction of movement in the plane when $\frac{3}{5} < h < 1$.
- (b) Construct the phase plane, including all nullclines, equilibria, and arrows indicating the direction of movement in the plane when $\frac{1}{10} < h < \frac{3}{5}$.
- (c) Construct the phase plane, including all nullclines, equilibria, and arrows indicating the direction of movement in the plane when $0 < h < \frac{1}{10}$.
- (d) From your results to parts (a), (b), and (c), determine how habitat destruction is expected to affect the coexistence of the two species.

Source: Adapted from S. Nee et al., “Dynamics of Metapopulations: Habitat Destruction and Competitive Coexistence,” *Journal of Animal Ecology* 61 (1992): 37–40.

25. Cell cycle dynamics The process of cell division is periodic, with repeated growth and division phases as the cell population multiplies. It has been suggested that the

division phase is triggered by high concentrations of a molecule called MPF (maturation promoting factor). The production of this factor is stimulated by another molecule called cyclin, and MPF eventually inhibits its own production. Using M and C to denote the concentrations of these two biomolecules (in mg/mL), a simple model for their interaction is

$$\frac{dM}{dt} = \alpha C + \beta CM^2 - \frac{\gamma M}{1 + M}$$

$$\frac{dC}{dt} = \delta - M$$

- (a) Suppose that $\alpha = 2$, $\beta = 1$, $\gamma = 10$, and $\delta = 1$. Construct the phase plane, including all nullclines, equilibria, and arrows indicating the direction of movement in the plane.
- (b) From your answer to part (a), what is the qualitative nature of the dynamics of M predicted by this model? What does this predict about the dynamics of cell division?
- (c) For any equilibrium found in part (a), specify whether it is locally stable, unstable, or if the information is inconclusive.

Source: Adapted from R. Norel et al., “A Model for the Adjustment of the Mitotic Clock by Cyclin and MPF Levels,” *Science* 251 (1991): 1076–78.

CASE STUDY 2c Hosts, Parasites, and Time-Travel



In this part of the case study you will formulate a mathematical model for the antagonistic interactions between *Daphnia* and its parasite using differential equations. Let’s suppose that there are two possible host genotypes (A and a) and two possible parasite genotypes (B and b). Parasites of type B can infect only hosts of type A, while parasites of type b can infect only hosts of type a (see Table 1). We will derive a set of two coupled differential equations that model the dynamics of the frequency of A in the host population and B in the parasite population.

Table 1
The outcome of challenges between different host and parasite genotypes.

	Host A	Host a
Parasite B	Infection occurs	Infection does not occur
Parasite b	Infection does not occur	Infection occurs

A common differential equation used in biology to model the frequency dynamics of a particular genotype is

$$\text{BB} \quad (1) \quad \frac{df}{dt} = f(1 - f)(r_1 - r_2)$$

where f is the frequency of type 1, and r_1 and r_2 are the per capita reproduction rates of the two types. For example, see Exercise 7.2.16. We will use an equation of this form for both the host and the parasite populations.

Suppose the per capita reproduction rate of uninfected hosts is r_q and that for infected hosts is $r_q - s_q$. The constant s_q is assumed to satisfy the inequality $0 < s_q < r_q$ and represents the reduction in reproductive output of a host due to infection. Similarly, the per capita reproduction rate of a parasite that is able to infect a host is r_p and that for one unable to infect a host is $r_p - s_p$ (the parasite can reproduce in the absence of the host, but it does so less well). The constant s_p is assumed to satisfy the inequality $0 < s_p < r_p$ and represents the reduction in reproductive output of a parasite if it is unable to infect a host.

Let's use q to denote the frequency of type A individuals in the host population and p to denote the frequency of type B individuals in the parasite population. Suppose that host-parasite encounters occur at random with respect to genotype.

1. With random encounters, the average per capita reproduction rate for hosts of a given type is $r_B p + r_b(1 - p)$, where r_B and r_b are the reproduction rates of the host when encountering a type B or type b parasite, respectively. Show that the average per capita reproduction rates of hosts of type A and a are therefore

$$\text{type A:} \quad r_q - p s_q$$

$$\text{type a:} \quad r_q - (1 - p) s_q$$

2. With random encounters, the average per capita reproduction rate for parasites of a given type is $r_A q + r_a(1 - q)$, where r_A and r_a are the reproduction rates of the parasite when encountering a type A or type a host, respectively. Show that the average per capita reproduction rates of parasites of type B and b are therefore

$$\text{type B:} \quad r_p - (1 - q) s_p$$

$$\text{type b:} \quad r_p - q s_p$$

3. Suppose both q and p satisfy differential equations of the form given in Equation 1. Show that q and p therefore satisfy

$$\frac{dq}{dt} = s_q q(1 - q)(1 - 2p)$$

$$\frac{dp}{dt} = s_p p(1 - p)(2q - 1)$$

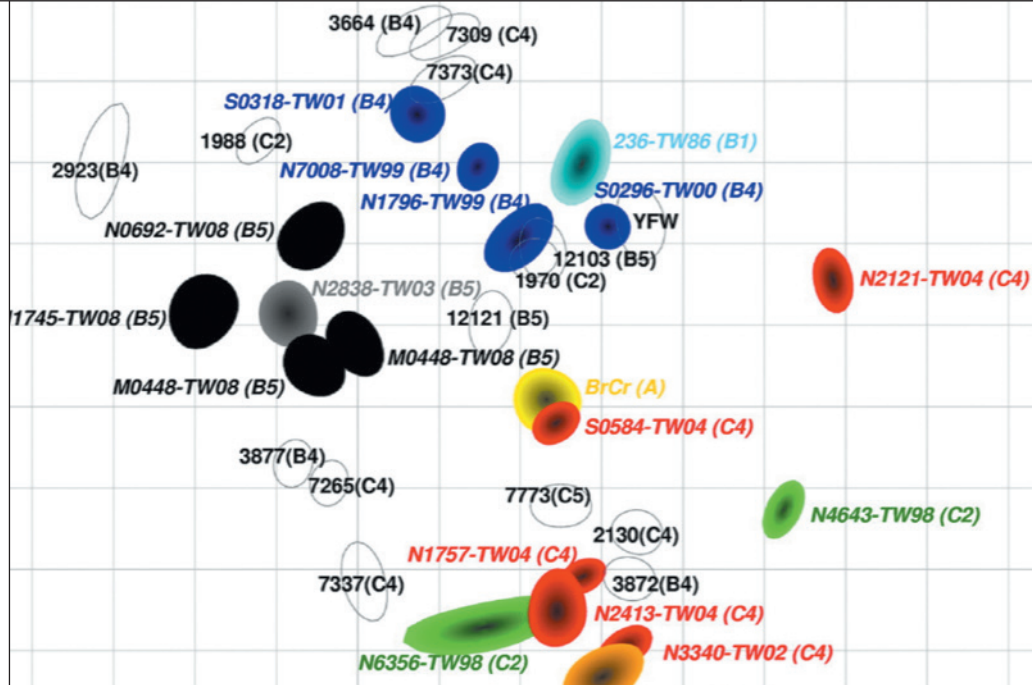
4. Construct the phase plane including all nullclines, equilibria, and arrows indicating the direction of movement in the plane.
5. Explain, qualitatively, how the frequencies of the two parasite genotypes are predicted to change over time. Similarly, explain how the frequencies of the two host genotypes are predicted to change over time.

Vectors and Matrix Models

8

Shown is an antigenic map of isolates of human enterovirus. Similar maps are constructed in Examples 8.1.3, 8.1.6, and 8.1.8.

Source: S.-W. Huang et al., "Reemergence of Enterovirus 71 in 2008 in Taiwan: Dynamics of Genetic and Antigenic Evolution from 1998 to 2008," *Journal of Clinical Microbiology* 47 (2009): 3653–62.



8.1 Coordinate Systems

8.2 Vectors

8.3 The Dot Product

PROJECT: Microarray Analysis of Genome Expression

PROJECT: Vaccine Escape

8.4 Matrix Algebra

8.5 Matrices and the Dynamics of Vectors

8.6 The Inverse and Determinant of a Matrix

PROJECT: Cubic Splines

8.7 Eigenvectors and Eigenvalues

8.8 Iterated Matrix Models

PROJECT: The Emergence of Geometric Order in Proliferating Cells

IN THIS CHAPTER WE INTRODUCE coordinate systems and vectors for three-dimensional space and higher. This will lead us into important ideas in linear algebra and the powerful techniques of matrix models. It also sets the stage for a more complete analysis of systems of differential equations and the study of functions of multiple variables.

8.1 Coordinate Systems

To locate a point in a plane, two numbers are necessary. We know that any point in the plane can be represented as an ordered pair (a, b) of real numbers, where a is the x -coordinate and b is the y -coordinate. For this reason, a plane is called two-dimensional.

Three-Dimensional Space

To locate a point in space, three numbers are required. We represent any point in space by an ordered triple (a, b, c) of real numbers. In order to do so we first choose a fixed point O (the origin) and three directed lines through O that are perpendicular to each other, called the **coordinate axes** and labeled the x -axis, y -axis, and z -axis. Usually we think of the x - and y -axes as being horizontal and the z -axis as being vertical, and we draw the orientation of the axes as in Figure 1. The direction of the z -axis is determined by the **right-hand rule** as illustrated in Figure 2: If you curl the fingers of your right hand around the z -axis in the direction of a 90° counterclockwise rotation from the positive x -axis to the positive y -axis, then your thumb points in the positive direction of the z -axis.

The three coordinate axes determine the three **coordinate planes** illustrated in Figure 3(a). The xy -plane is the plane that contains the x - and y -axes; the yz -plane contains the y - and z -axes; the xz -plane contains the x - and z -axes. These three coordinate planes divide space into eight parts, called **octants**. The **first octant**, in the foreground, is determined by the positive axes.

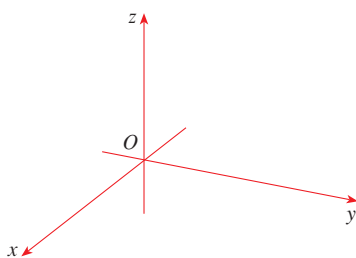


FIGURE 1
Coordinate axes

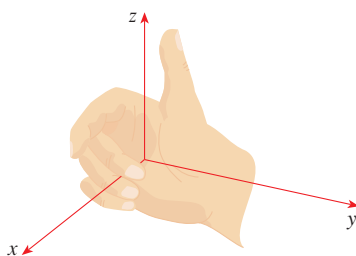


FIGURE 2
Right-hand rule

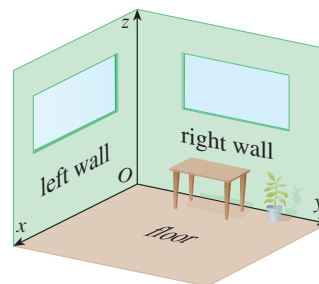
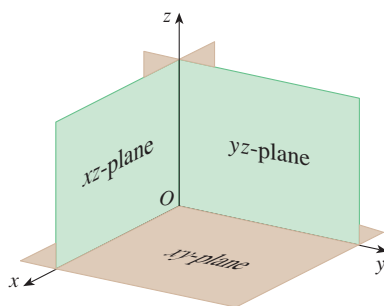


FIGURE 3

(a) Coordinate planes

(b)

Because many people have some difficulty visualizing diagrams of three-dimensional figures, you may find it helpful to do the following [see Figure 3(b)]. Look at any bottom corner of a room and call the corner the origin. The wall on your left is in the xz -plane, the wall on your right is in the yz -plane, and the floor is in the xy -plane. The x -axis runs along the intersection of the floor and the left wall. The y -axis runs along the intersection of the floor and the right wall. The z -axis runs up from the floor toward the ceiling along the intersection of the two walls. You are situated in the first octant, and you can now imagine

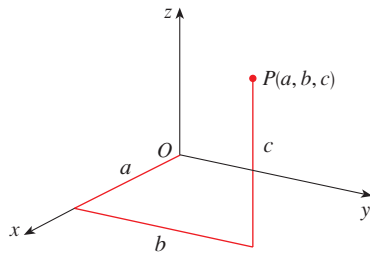


FIGURE 4

seven other rooms situated in the other seven octants (three on the same floor and four on the floor below), all connected by the common corner point O .

Now if P is any point in space, let a be the (directed) distance from the yz -plane to P , let b be the distance from the xz -plane to P , and let c be the distance from the xy -plane to P . We represent the point P by the ordered triple (a, b, c) of real numbers and we call a , b , and c the **coordinates** of P ; a is the x -coordinate, b is the y -coordinate, and c is the z -coordinate. Thus, to locate the point (a, b, c) , we can start at the origin O and move a units along the x -axis, then b units parallel to the y -axis, and then c units parallel to the z -axis as in Figure 4.

The point $P(a, b, c)$ determines a rectangular box as in Figure 5. If we drop a perpendicular from P to the xy -plane, we get a point Q with coordinates $(a, b, 0)$ called the **projection** of P onto the xy -plane. Similarly, $R(0, b, c)$ and $S(a, 0, c)$ are the projections of P onto the yz -plane and xz -plane, respectively.

As numerical illustrations, the points $(-4, 3, -5)$ and $(3, -2, -6)$ are plotted in Figure 6.

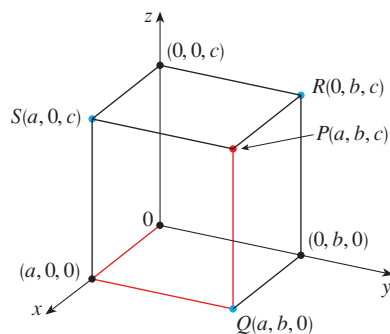


FIGURE 5

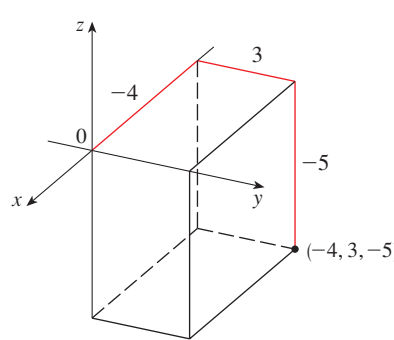


FIGURE 6

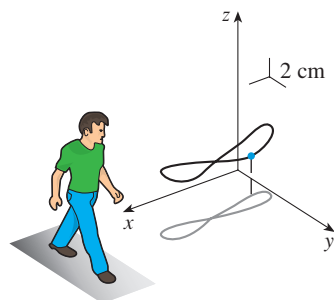
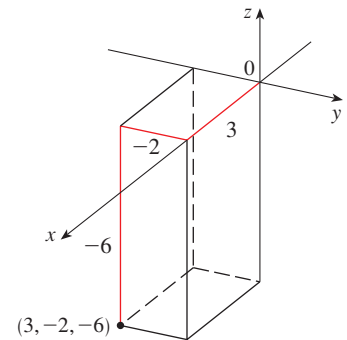


FIGURE 7

The blue dot indicates the position at a fixed time. The gray curve is a projection of the black curve onto the horizontal plane. (Image of person is not to scale.)

EXAMPLE 1 | Human biomechanics The position of the center of a human walking on a treadmill at any point in time is described by its x , y , and z coordinates (see Figure 7). Over time the point moves and traces out a curve in space. Which directions of movement are revealed by the projection of this point onto the horizontal coordinate plane? Explain why the curve projected onto this plane has the shape shown in Figure 7.

SOLUTION The projection onto the horizontal coordinate plane shows motion from side to side as well as from front to back. It does not reveal any motion in the vertical direction. The projected curve shown in the figure resembles a “figure eight” that is elongated in the direction of side-to-side movement (the x -direction). People sway from side to side with each step while walking, causing the movement in the x -direction. At the same time, each step that the person takes moves him forward slightly, and the turning of the treadmill then moves him backward again. Together these movements trace out a figure eight in the horizontal plane. In this example, the magnitude of the side-to-side movement is larger than the magnitude of the movement front to back, which causes the projected curve to appear elongated in the x -direction. ■

The Cartesian product $\mathbb{R} \times \mathbb{R} \times \mathbb{R} = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$ is the set of all ordered triples of real numbers and is denoted by $\mathbb{R}^3$. We have given a one-to-one correspondence between points P in space and ordered triples (a, b, c) in $\mathbb{R}^3$. It is called a **three-dimensional rectangular coordinate system**. Notice that, in terms of

coordinates, the first octant can be described as the set of points whose coordinates are all positive.

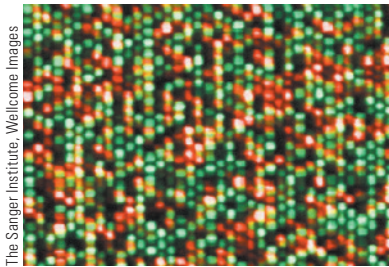


FIGURE 8 Expression level of each gene is measured as color intensity of a dot on a microarray, as shown here. The entire microarray displays the genome expression profile. See also the Project on page 513.

EXAMPLE 2 | BB Genome expression profiles Due to advances in biotechnology, researchers can now quantify gene expression across the entire genome of organisms in response to various experimental perturbations. The set of the levels of expression of a collection of genes is called a **genome expression profile** (see Figure 8). Most perturbations result in some genes being upregulated (positive expression) and others being suppressed (negative expression). Consider three hypothetical genes, A, B, and C, and suppose two different experiments are conducted that result in the following dimensionless expression profiles: Experiment 1 $(A, B, C) = (1, 1, -1.5)$ and Experiment 2 $(A, B, C) = (1.5, -1, 0.2)$. Plot these data points in $\mathbb{R}^3$. Interpret their projections on each of the three coordinate planes.

SOLUTION Using the x -, y -, and z -coordinates for the expression levels of genes A, B, and C, respectively, we obtain Figure 9(a) for Experiment 1 and Figure 9(b) for Experiment 2. The projections of each data point on the three different coordinate planes reveal the various possible two-gene expression profiles.

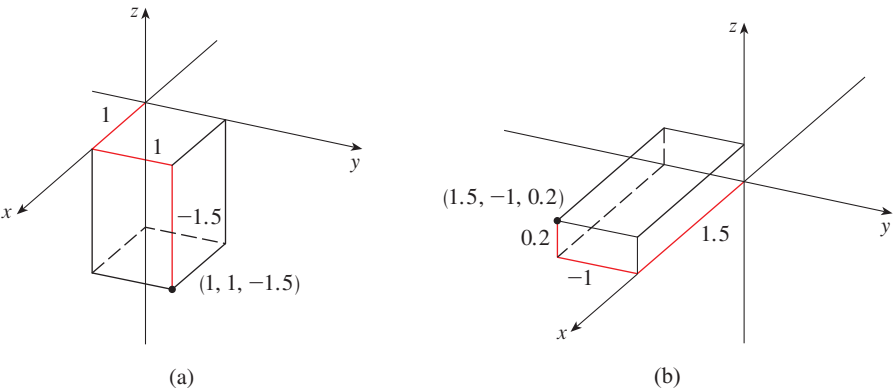


FIGURE 9

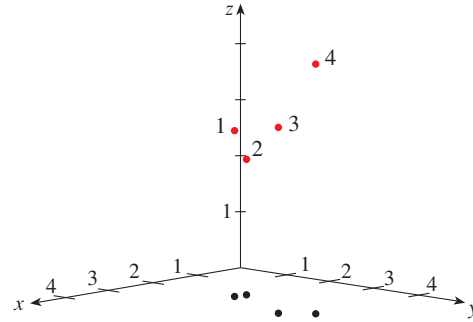
EXAMPLE 3 | BB Antigenic cartography The extent to which viruses react with the immune system is sometimes measured by testing their ability to bind with a large panel of different immune molecules called antisera. Suppose the binding ability of four influenza strains with three different kinds of antiserum molecules are given by the dimensionless numbers in the following table.

	Antiserum 1	Antiserum 2	Antiserum 3
Strain 1	2.06	1.92	2.96
Strain 2	1.79	1.91	2.44
Strain 3	2.68	3.53	3.31
Strain 4	2.39	4.05	4.46

Plot the data for the four strains in three-dimensional space. This space is referred to as **antigenic space** and construction of such an antigenicity plot is referred to as **antigenic cartography**.

SOLUTION Using the x -, y -, and z -coordinates for the binding ability to antisera 1, 2, and 3, we obtain Figure 10. The binding ability to any pair of antisera is given by the projection of these four red points onto the various coordinate planes.

FIGURE 10
Four strains plotted in three-dimensional antigenic space (red dots), as well as the projection onto the two-dimensional antigenic space of antisera 1 and 2 (black dots).



In two-dimensional analytic geometry, the graph of an equation involving x and y is a curve in $\mathbb{R}^2$. In three-dimensional analytic geometry, an equation in x , y , and z represents a *surface* in $\mathbb{R}^3$.

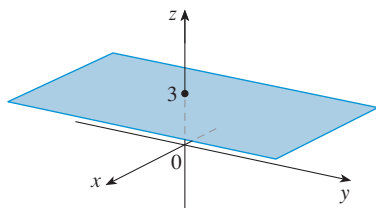
EXAMPLE 4 | What surfaces in $\mathbb{R}^3$ are represented by the following equations?

(a) $z = 3$

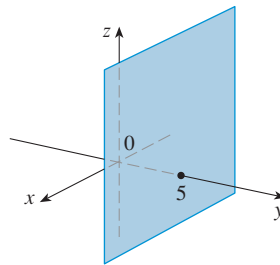
(b) $y = 5$

SOLUTION

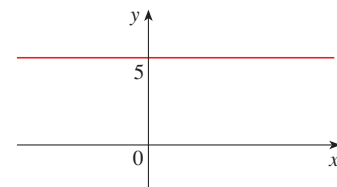
(a) The equation $z = 3$ represents the set $\{(x, y, z) \mid z = 3\}$, which is the set of all points in $\mathbb{R}^3$ whose z -coordinate is 3. This is the horizontal plane that is parallel to the xy -plane and three units above it as in Figure 11(a).



(a) $z = 3$, a plane in $\mathbb{R}^3$



(b) $y = 5$, a plane in $\mathbb{R}^3$



(c) $y = 5$, a line in $\mathbb{R}^2$

FIGURE 11

(b) The equation $y = 5$ represents the set of all points in $\mathbb{R}^3$ whose y -coordinate is 5. This is the vertical plane that is parallel to the xz -plane and five units to the right of it as in Figure 11(b).

Note that when an equation is given, we must understand from the context whether it represents a curve in $\mathbb{R}^2$ or a surface in $\mathbb{R}^3$. For example, $y = 5$ represents a plane in $\mathbb{R}^3$, but of course $y = 5$ can also represent a line in $\mathbb{R}^2$ if we are dealing with two-dimensional analytic geometry. See Figure 11(b) and (c).

The familiar formula for the distance between two points in a plane is easily extended to the following three-dimensional formula.

Distance Formula in Three Dimensions The distance $|P_1P_2|$ between the points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is

$$|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

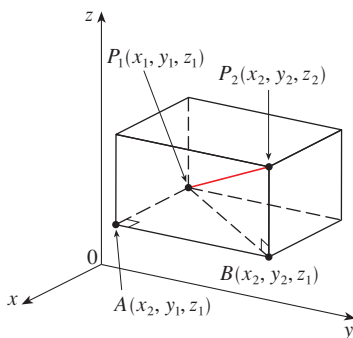


FIGURE 12

To see why this formula is true, we construct a rectangular box as in Figure 12, where P_1 and P_2 are opposite vertices and the faces of the box are parallel to the coordinate planes. If $A(x_2, y_1, z_1)$ and $B(x_2, y_2, z_1)$ are the vertices of the box indicated in the figure, then

$$|P_1A| = |x_2 - x_1| \quad |AB| = |y_2 - y_1| \quad |BP_2| = |z_2 - z_1|$$

Because triangles P_1BP_2 and P_1AB are both right-angled, two applications of the Pythagorean Theorem give

$$|P_1P_2|^2 = |P_1B|^2 + |BP_2|^2$$

and

$$|P_1B|^2 = |P_1A|^2 + |AB|^2$$

Combining these equations, we get

$$\begin{aligned} |P_1P_2|^2 &= |P_1A|^2 + |AB|^2 + |BP_2|^2 \\ &= |x_2 - x_1|^2 + |y_2 - y_1|^2 + |z_2 - z_1|^2 \\ &= (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2 \end{aligned}$$

Therefore $|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

EXAMPLE 5 | The distance from the point $P(2, -1, 7)$ to the point $Q(1, -3, 5)$ is

$$|PQ| = \sqrt{(1 - 2)^2 + (-3 + 1)^2 + (5 - 7)^2} = \sqrt{1 + 4 + 4} = 3$$

EXAMPLE 6 | **BB** Antigenic cartography (continued) What is the antigenic distance between influenza strain 2 and strain 4 in Example 3?

SOLUTION The distance between strain 2 and strain 4 is the distance between point $P(1.79, 1.91, 2.44)$ and point $Q(2.39, 4.05, 4.46)$. This is

$$|PQ| = \sqrt{(2.39 - 1.79)^2 + (4.05 - 1.91)^2 + (4.46 - 2.44)^2} \approx 3$$

EXAMPLE 7 | Find an equation of a sphere with radius r and center $C(h, k, l)$.

SOLUTION By definition, a sphere is the set of all points $P(x, y, z)$ whose distance from C is r . (See Figure 13.) Thus P is on the sphere if and only if $|PC| = r$. Squaring both sides, we have $|PC|^2 = r^2$ or

$$(x - h)^2 + (y - k)^2 + (z - l)^2 = r^2$$

The result of Example 7 is worth remembering.

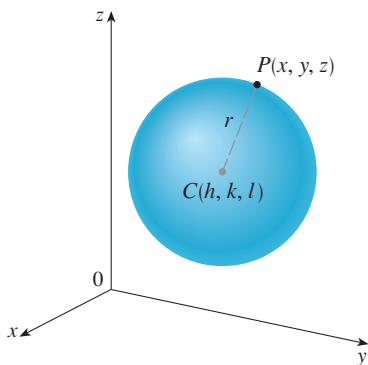


FIGURE 13

Equation of a Sphere An equation of a sphere with center $C(h, k, l)$ and radius r is

$$(x - h)^2 + (y - k)^2 + (z - l)^2 = r^2$$

In particular, if the center is the origin O , then an equation of the sphere is

$$x^2 + y^2 + z^2 = r^2$$

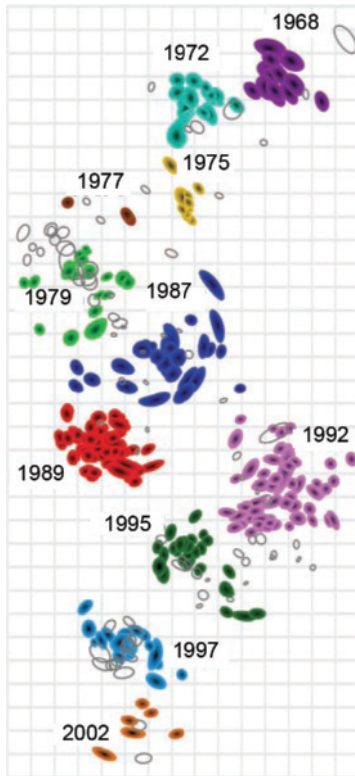


FIGURE 14

Higher-Dimensional Space

Just as we extended two-dimensional space to three-dimensional space, we can go further and generalize to n -dimensional space. Although we cannot visualize such spaces, we can still work with them mathematically. This is extremely important in the life sciences because the systems and objects that we seek to describe mathematically are often characterized by several variables. The Cartesian product $\mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}$, where the product involves n copies of $\mathbb{R}$, is denoted $\mathbb{R}^n$ and is defined as $\mathbb{R}^n = \{(x_1, x_2, \dots, x_n) \mid x_1, x_2, \dots, x_n \in \mathbb{R}\}$. This is the set of all ordered n -tuples of real numbers.

Distance Formula in n Dimensions The distance $|P_1P_2|$ between the points $P_1(a_1, \dots, a_n)$ and $P_2(b_1, \dots, b_n)$ is

$$|P_1P_2| = \sqrt{(b_1 - a_1)^2 + \cdots + (b_n - a_n)^2}$$

EXAMPLE 8 | BB Antigenic cartography (continued) Antigenic cartography has been carried out for 273 human influenza viruses collected over time from 1968 through to 2002.¹ Each virus was tested for its binding ability to 79 different antisera, producing a 79-dimensional antigenic space. The result is a plot of 273 points in $\mathbb{R}^{79}$. We cannot visualize such a high-dimensional space but, just as in three-dimensional space, we can project the points onto lower-dimensional surfaces. Although the way in which this is done can be complicated, Figure 14 gives an example of these data projected onto a two-dimensional plane. Despite the virus data existing in 79-dimensional antigenic space, we get a remarkably good visualization of the viruses, and how they have changed antigenically over a 34-year period, by projecting the points onto a plane representing a two-dimensional antigenic space.

EXERCISES 8.1

- Suppose you start at the origin, move along the x -axis a distance of 4 units in the positive direction, and then move downward a distance of 3 units. What are the coordinates of your position?
- Sketch the points $(0, 5, 2)$, $(4, 0, -1)$, $(2, 4, 6)$, and $(1, -1, 2)$ on a single set of coordinate axes.
- Which of the points $P(6, 2, 3)$, $Q(-5, -1, 4)$, and $R(0, 3, 8)$ is closest to the xz -plane? Which point lies in the yz -plane?
- What are the projections of the point $(2, 3, 5)$ on the xy -, yz -, and xz -planes? Draw a rectangular box with the origin and $(2, 3, 5)$ as opposite vertices and with its faces parallel to the coordinate planes. Label all vertices of the box. Find the length of the diagonal of the box.
- Describe and sketch the surface in $\mathbb{R}^3$ represented by the equation $x + y = 2$.
- (a) What does the equation $x = 4$ represent in $\mathbb{R}^2$? What does it represent in $\mathbb{R}^3$? Illustrate with sketches.
(b) What does the equation $y = 3$ represent in $\mathbb{R}^3$? What does $z = 5$ represent? What does the pair of equations $y = 3, z = 5$ represent? In other words, describe the set of points (x, y, z) such that $y = 3$ and $z = 5$. Illustrate with a sketch.
- Find the lengths of the sides of the triangle PQR . Is it a right triangle? Is it an isosceles triangle?
(a) $P(3, -2, -3)$, $Q(7, 0, 1)$, $R(1, 2, 1)$
(b) $P(2, -1, 0)$, $Q(4, 1, 1)$, $R(4, -5, 4)$
- Find the distance from $(3, 7, -5)$ to each of the following.
(a) The xy -plane
(b) The yz -plane
(c) The xz -plane
(d) The x -axis
(e) The y -axis
(f) The z -axis

1. D. Smith et al., "Mapping the Antigenic and Genetic Evolution of Influenza Virus," *Science* 305 (2004): 371–76.

9. Determine whether the points lie on a straight line.
 (a) $A(2, 4, 2)$, $B(3, 7, -2)$, $C(1, 3, 3)$
 (b) $D(0, -5, 5)$, $E(1, -2, 4)$, $F(3, 4, 2)$
10. Find an equation of the sphere with center $(2, -6, 4)$ and radius 5. Describe its intersection with each of the coordinate planes.
11. Find an equation of the sphere that passes through the point $(4, 3, -1)$ and has center $(3, 8, 1)$.
12. Find an equation of the sphere that passes through the origin and whose center is $(1, 2, 3)$.

13–16 Show that the equation represents a sphere, and find its center and radius.

13. $x^2 + y^2 + z^2 - 6x + 4y - 2z = 11$

14. $x^2 + y^2 + z^2 + 8x - 6y + 2z + 17 = 0$

15. $2x^2 + 2y^2 + 2z^2 = 8x - 24z + 1$

16. $3x^2 + 3y^2 + 3z^2 = 10 + 6y + 12z$

17. (a) Prove that the midpoint of the line segment from $P_1(x_1, y_1, z_1)$ to $P_2(x_2, y_2, z_2)$ is

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$$

- (b) A median of a triangle is a line segment joining a vertex to the midpoint of the opposite side. Find the lengths of the medians of the triangle with vertices $A(1, 2, 3)$, $B(-2, 0, 5)$, and $C(4, 1, 5)$.
18. Find an equation of a sphere if one of its diameters has endpoints $(2, 1, 4)$ and $(4, 3, 10)$.
19. Find equations of the spheres with center $(2, -3, 6)$ that touch (a) the xy -plane, (b) the yz -plane, (c) the xz -plane.
20. Find an equation of the largest sphere with center $(5, 4, 9)$ that is contained in the first octant.

21–30 Describe in words the region of $\mathbb{R}^3$ represented by the equations or inequalities.

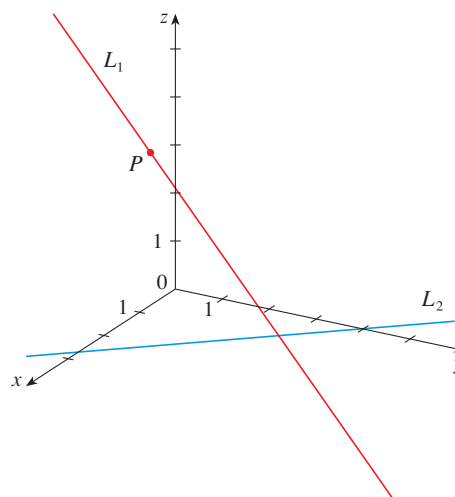
21. $x = 5$ 22. $y = -2$
 23. $y < 8$ 24. $x \geq -3$
 25. $0 \leq z \leq 6$ 26. $z^2 = 1$
 27. $x^2 + y^2 = 4$, $z = -1$ 28. $y^2 + z^2 = 16$
 29. $x^2 + y^2 + z^2 \leq 3$ 30. $x = z$

31–34 Write inequalities to describe the region.

31. The region between the yz -plane and the vertical plane $x = 5$
32. The solid cylinder that lies on or below the plane $z = 8$ and on or above the disk in the xy -plane with center the origin and radius 2

33. The region consisting of all points between (but not on) the spheres of radius r and R centered at the origin, where $r < R$
34. The solid upper hemisphere of the sphere of radius 2 centered at the origin

35. The figure shows a line L_1 in space and a second line L_2 , which is the projection of L_1 onto the xy -plane. (In other words, the points on L_2 are directly beneath, or above, the points on L_1 .)
 (a) Find the coordinates of the point P on the line L_1 .
 (b) Locate on the diagram the points A , B , and C , where the line L_1 intersects the xy -plane, the yz -plane, and the xz -plane, respectively.



36. **Darwin's finches** have been used to study how differences in bird morphology are related to differences in diet. Morphological measurements (in mm) of three species are given in the table for three traits.

Species	Wing length	Tarsus length	Beak length
<i>G. difficilis</i>	64	18.1	9.6
<i>G. fuliginosa</i>	62.1	17.9	8.6
<i>G. scandens</i>	73.1	21.1	14.5

The proportion of time spent feeding on different types of food for these three species is given in the following table.

Species	Seeds	Pollen	Other
<i>G. difficilis</i>	0.67	0.23	0.1
<i>G. fuliginosa</i>	0.7	0.28	0.02
<i>G. scandens</i>	0.14	0	0.86

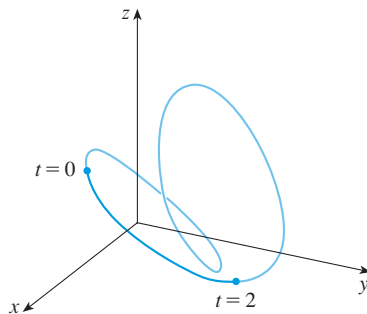
- (a) Thinking of the morphology of each species as a point in $\mathbb{R}^3$, calculate the morphological distance between each pair of species.
- (b) Thinking of the diet of each species as a point in $\mathbb{R}^3$, calculate the diet distance between each pair of species.

- (c) Do species that are morphologically most similar also tend to have the most similar diets?

Source: Adapted from D. Schluter et al., "Ecological Correlates of Morphological Evolution in a Darwin's Finch, *Geospiza difficilis*," *Evolution* 38 (1984): 856–69.

- 37. Human biomechanics** The trajectory of the center of a human walking on a treadmill is shown in the figure. Here x denotes the lateral position, y the position forward or backward, and z the vertical position. The coordinates at time $t = 0$ are $(1.2, 0, 1.5)$ and at $t = 2$ are $(1, 2, 0.5)$, where distances are measured in cm.

- (a) What is magnitude of the the net lateral distance traveled over the first two seconds?
- (b) What is the magnitude of the net distance traveled in the vertical direction over the first two seconds?
- (c) What is the net distance traveled through three-dimensional space over the first two seconds?

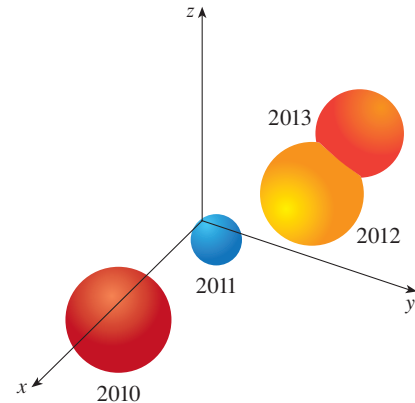


Source: Adapted from L. Tesio et al., "The 3D Trajectory of the Body Centre of Mass during Adult Human Walking: Evidence for a Speed-Curvature Power Law," *Journal of Biomechanics* 44 (2011): 732–40.

- 38. Vaccine design** Most vaccines protect only against pathogens that fall within a certain region of antigenic space. Suppose that a vaccine protects against any strain falling within a sphere of radius 2 centered at the point $(2, 1, 0)$ in antigenic space. For each strain, determine whether this vaccine will be effective.
- (a) A strain located at $(0, 0, 0)$ in antigenic space
- (b) A strain located at $(1, 0, 3)$ in antigenic space
- (c) A strain located at $(1, 0, 1)$ in antigenic space
- (d) A strain located at $(1/4, 2, 1)$ in antigenic space

- 39. Antigenic evolution and vaccination** Antigenic data like those in Figure 14 can be summarized by taking the points from each year and drawing the smallest possible circle that encompasses these data. This results in a temporal sequence of circles in antigenic space, one for each

year. Similarly, if the antigenic data are in three dimensions, spheres can be drawn around the data for each year, as shown in the figure. If the circles (or spheres) from years x and $x + 1$ overlap, then the amount of antigenic change between these years is relatively small. In such cases we might expect a single vaccine to work for both years. If the circles or spheres do not overlap, then we might need different vaccines for each year.



- (a) Suppose the antigenic data are two-dimensional, and the circles for two successive years are given by the equations $(x - 2)^2 + (y - 3)^2 = 1$ and $(x - 3)^2 + (y - 2)^2 = \frac{1}{4}$. Would a single vaccine work for both years?
- (b) Suppose the antigenic data are three-dimensional, and the spheres for two successive years are given by the equations

$$(x - 2)^2 + (y - 3)^2 + (z - 1)^2 = 1$$

and $(x - 3)^2 + (y - 2)^2 + z^2 = \frac{1}{4}$

Would a single vaccine work for both years?

- (c) Notice that the x - and y -coordinates of the centers of the circles in part (a) are the same as the x - and y -coordinates of the centers of the spheres in part (b), and the radii are the same as well. What is the relationship between the plot of the circles in part (a) and the plot of the spheres in part (b)?

- 40.** Describe and sketch a solid with the following properties: When illuminated by rays parallel to the z -axis, its shadow is a circular disk. If the rays are parallel to the y -axis, its shadow is a square. If the rays are parallel to the x -axis, its shadow is an isosceles triangle.

8.2 Vectors

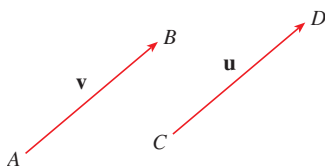


FIGURE 1
Equivalent vectors

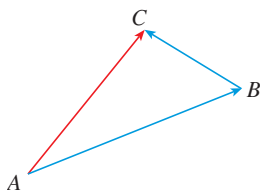


FIGURE 2

The term **vector** is used by scientists to indicate a quantity (such as displacement or velocity or force) that has both magnitude and direction. A vector is often represented by an arrow or a directed line segment. The length of the arrow represents the magnitude of the vector and the arrow points in the direction of the vector. We denote a vector by printing a letter in lowercase boldface ($\mathbf{v}$) or by putting an arrow above the letter ($\vec{v}$).

For instance, suppose a particle moves along a line segment from point A to point B . The corresponding **displacement vector** $\mathbf{v}$, shown in Figure 1, has **initial point** A (the tail) and **terminal point** B (the tip) and we indicate this by writing $\mathbf{v} = \overrightarrow{AB}$. Notice that the vector $\mathbf{u} = \overrightarrow{CD}$ has the same length and the same direction as $\mathbf{v}$ even though it is in a different position. We say that $\mathbf{u}$ and $\mathbf{v}$ are **equivalent** (or **equal**) and we write $\mathbf{u} = \mathbf{v}$. The **zero vector**, denoted by $\mathbf{0}$, has length 0. It is the only vector with no specific direction.

Combining Vectors

Suppose a particle moves from A to B , so its displacement vector is $\overrightarrow{AB}$. Then the particle changes direction and moves from B to C , with displacement vector $\overrightarrow{BC}$ as in Figure 2. The combined effect of these displacements is that the particle has moved from A to C . The resulting displacement vector $\overrightarrow{AC}$ is called the *sum* of $\overrightarrow{AB}$ and $\overrightarrow{BC}$ and we write

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$$

In general, if we start with vectors $\mathbf{u}$ and $\mathbf{v}$, we first move $\mathbf{v}$ so that its tail coincides with the tip of $\mathbf{u}$ and define the sum of $\mathbf{u}$ and $\mathbf{v}$ as follows.

Definition of Vector Addition If $\mathbf{u}$ and $\mathbf{v}$ are vectors positioned so the initial point of $\mathbf{v}$ is at the terminal point of $\mathbf{u}$, then the sum $\mathbf{u} + \mathbf{v}$ is the vector from the initial point of $\mathbf{u}$ to the terminal point of $\mathbf{v}$.

The definition of vector addition is illustrated in Figure 3. You can see why this definition is sometimes called the **Triangle Law**.

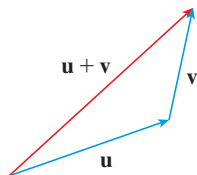


FIGURE 3
The Triangle Law

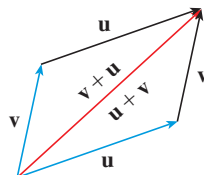


FIGURE 4
The Parallelogram Law

In Figure 4 we start with the same vectors $\mathbf{u}$ and $\mathbf{v}$ as in Figure 3 and draw another copy of $\mathbf{v}$ with the same initial point as $\mathbf{u}$. Completing the parallelogram, we see that $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$. This also gives another way to construct the sum: If we place $\mathbf{u}$ and $\mathbf{v}$ so they start at the same point, then $\mathbf{u} + \mathbf{v}$ lies along the diagonal of the parallelogram with $\mathbf{u}$ and $\mathbf{v}$ as sides. (This is called the **Parallelogram Law**.)

EXAMPLE 1 | **BB** **Antigenic cartography** The evolution of influenza in antigenic space across years can be viewed as a vector, since it is characterized by both its magnitude and direction of change (see also Example 8.1.8). Figure 5 shows clusters of influenza virus in two-dimensional antigenic space from several years. Illustrate how the vector of antigenic change between the centers of the viral clusters in 1989 and 1995 is the sum of the corresponding vectors from 1989–1992 and 1992–1995. Use both the Triangle and the Parallelogram Law.

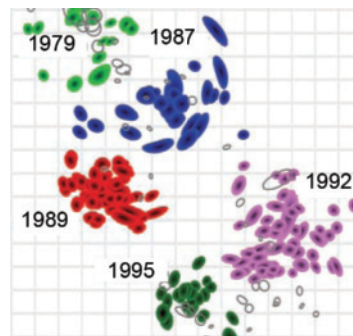


FIGURE 5

SOLUTION The vector corresponding to the antigenic change from 1989 to 1995 is labeled as $\mathbf{c}$ in Figure 6. For the Triangle Law, first we draw a vector from the 1989 cluster to the 1992 cluster (labeled $\mathbf{a}$). Then we draw a vector from the 1992 cluster to the 1995 cluster (labeled $\mathbf{b}$). We see that $\mathbf{a} + \mathbf{b} = \mathbf{c}$. Figure 6(a) illustrates the Triangle Law.

For the Parallelogram Law, we again draw a vector from the 1989 cluster to the 1992 cluster. Then we draw a vector from the 1992 cluster to the 1995 cluster and translate it so that it starts where $\mathbf{a}$ starts. The vector $\mathbf{c}$ then lies on the diagonal of the corresponding parallelogram, as shown in Figure 6(b).

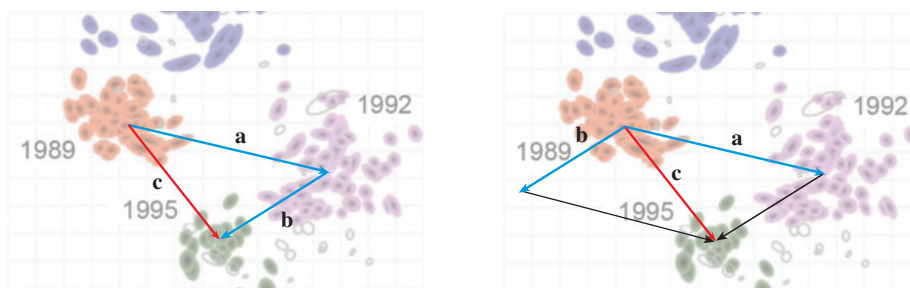


FIGURE 6

(a) The Triangle Law

(b) The Parallelogram Law

It is possible to multiply a vector by a real number c . (In this context we call the real number c a **scalar** to distinguish it from a vector.) For instance, we want $2\mathbf{v}$ to be the same vector as $\mathbf{v} + \mathbf{v}$, which has the same direction as $\mathbf{v}$ but is twice as long. In general, we multiply a vector by a scalar as follows.

Definition of Scalar Multiplication If c is a scalar and $\mathbf{v}$ is a vector, then the **scalar multiple** $c\mathbf{v}$ is the vector whose length is $|c|$ times the length of $\mathbf{v}$ and whose direction is the same as $\mathbf{v}$ if $c > 0$ and is opposite to $\mathbf{v}$ if $c < 0$. If $c = 0$ or $\mathbf{v} = \mathbf{0}$, then $c\mathbf{v} = \mathbf{0}$.

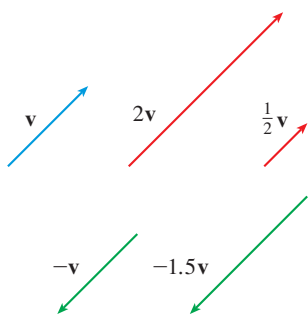


FIGURE 7
Scalar multiples of $\mathbf{v}$

This definition is illustrated in Figure 7. We see that real numbers work like scaling factors here; that's why we call them scalars. Notice that two nonzero vectors are **parallel** if they are scalar multiples of one another. In particular, the vector $-\mathbf{v} = (-1)\mathbf{v}$ has the same length as $\mathbf{v}$ but points in the opposite direction. We call it the **negative** of $\mathbf{v}$.

By the **difference** $\mathbf{u} - \mathbf{v}$ of two vectors we mean

$$\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v})$$

So we can construct $\mathbf{u} - \mathbf{v}$ by first drawing the negative of $\mathbf{v}$, $-\mathbf{v}$, and then adding it to $\mathbf{u}$ by the Parallelogram Law as in Figure 8(a). Alternatively, since $\mathbf{v} + (\mathbf{u} - \mathbf{v}) = \mathbf{u}$, the vector $\mathbf{u} - \mathbf{v}$, when added to $\mathbf{v}$, gives $\mathbf{u}$. So we could construct $\mathbf{u} - \mathbf{v}$ as in Figure 8(b) by means of the Triangle Law.

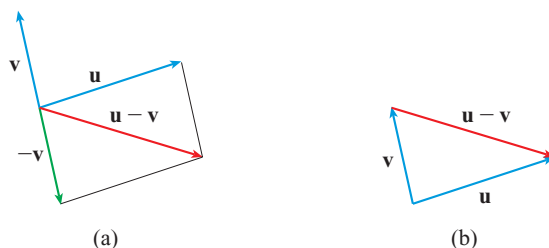


FIGURE 8
Drawing $\mathbf{u} - \mathbf{v}$

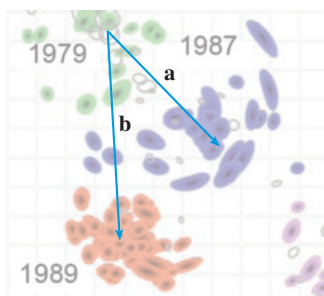


FIGURE 9

EXAMPLE 2 | **BB** Antigenic cartography (continued) Using the vectors $\mathbf{a}$ and $\mathbf{b}$ in Figure 9, illustrate the vector of antigenic change $\mathbf{b} - \mathbf{a}$ with both the Triangle and the Parallelogram Laws.

SOLUTION For the Triangle Law we seek the vector that, when added to $\mathbf{a}$ gives $\mathbf{b}$. This is obtained as the vector from the terminal point of $\mathbf{a}$ to the terminal point of $\mathbf{b}$, as shown in Figure 10(a).

For the Parallelogram Law we first construct the vector $-\mathbf{a}$. Then we add it to $\mathbf{b}$ using the Parallelogram Law as in Figure 10(b). Although the vectors constructed in Figures 10(a) and 10(b) are in different positions, they are equivalent vectors because they represent the same direction and magnitude of antigenic change.

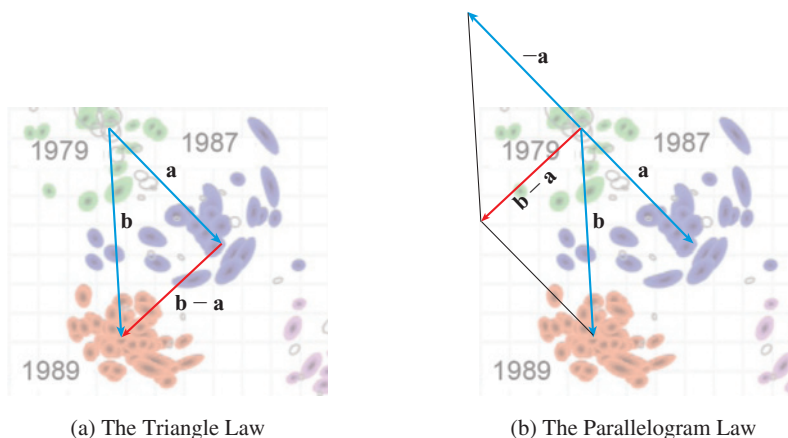


FIGURE 10

(a) The Triangle Law

(b) The Parallelogram Law

Components

For some purposes it's best to introduce a coordinate system and treat vectors algebraically. If we place the initial point of a vector $\mathbf{a}$ at the origin of a rectangular coordinate

system, then the terminal point of $\mathbf{a}$ has coordinates of the form (a_1, a_2) or (a_1, a_2, a_3) , depending on whether our coordinate system is two- or three-dimensional (see Figure 11).

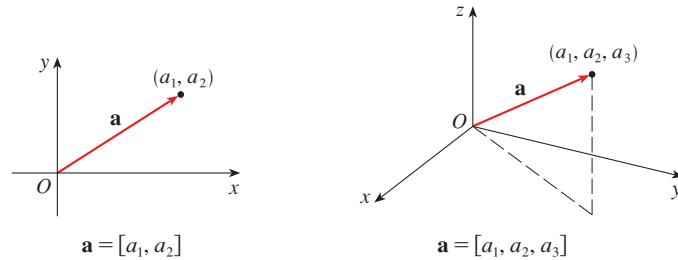


FIGURE 11

These coordinates are called the **components** of $\mathbf{a}$ and we write

$$\mathbf{a} = [a_1, a_2] \quad \text{or} \quad \mathbf{a} = [a_1, a_2, a_3]$$

The component form of a vector is sometimes written using angled brackets rather than square brackets. So, for instance, vector $\mathbf{a} = [a_1, a_2, a_3]$ can be equivalently written as $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$.

We use the notation $[a_1, a_2]$ for the ordered pair that refers to a vector so as not to confuse it with the ordered pair (a_1, a_2) that refers to a point in the plane.

For instance, the vectors shown in Figure 12 are all equivalent to the vector $\vec{OP} = [3, 2]$ whose terminal point is $P(3, 2)$. What they have in common is that the terminal point is reached from the initial point by a displacement of three units to the right and two upward. We can think of all these geometric vectors as **representations** of the algebraic vector $\mathbf{a} = [3, 2]$. The particular representation $\vec{OP}$ from the origin to the point $P(3, 2)$ is called the **position vector** of the point P .

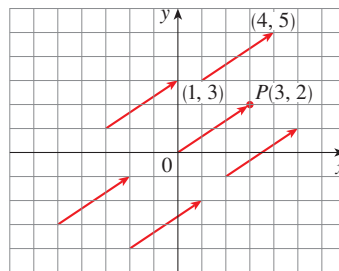


FIGURE 12

Representations of the vector $\mathbf{a} = [3, 2]$

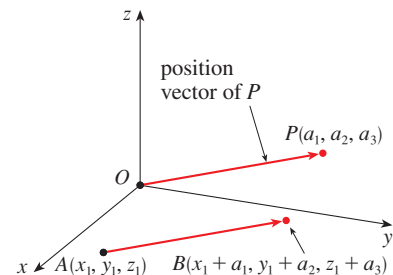


FIGURE 13

Representations of $\mathbf{a} = [a_1, a_2, a_3]$

In three dimensions, the vector $\mathbf{a} = \vec{OP} = [a_1, a_2, a_3]$ is the **position vector** of the point $P(a_1, a_2, a_3)$. (See Figure 13.) Let's consider any other representation $\vec{AB}$ of $\mathbf{a}$, where the initial point is $A(x_1, y_1, z_1)$ and the terminal point is $B(x_2, y_2, z_2)$. Then we must have $x_2 = x_1 + a_1$, $y_2 = y_1 + a_2$, and $z_2 = z_1 + a_3$ and so $a_1 = x_2 - x_1$, $a_2 = y_2 - y_1$, and $a_3 = z_2 - z_1$. Thus we have the following result.

(1) Given the points $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$, the vector $\mathbf{a}$ with representation $\vec{AB}$ is

$$\mathbf{a} = [x_2 - x_1, y_2 - y_1, z_2 - z_1]$$